CHAPTER 04

Analyzing Organic Reactions

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: ece9d543b87e8de535e4577e12ce3e6af00c3d6807faf0b3834f5e1761372839

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch04:exercise:pka-equilibrium | pka-equilibrium:stage-1, pka-equilibrium:stage-2, pka-equilibrium:stage-3

ch04:exercise:identify-partners | identify-partners:stage-1, identify-partners:stage-2, identify-partners:stage-3

ch04:exercise:sn2-rate | sn2-rate:stage-1, sn2-rate:stage-2, sn2-rate:stage-3

ch04:exercise:carbon-redox | carbon-redox:stage-1, carbon-redox:stage-2, carbon-redox:stage-3

ch04:exercise:selective-reduction | selective-reduction:stage-1, selective-reduction:stage-2, selective-reduction:stage-3

ch04:lesson:acids | acids:rule, acids:cue, acids:trap, acids:app, acids:step-1, acids:step-2

ch04:lesson:partners | partners:rule, partners:cue, partners:trap, partners:app, partners:step-1, partners:step-2

ch04:lesson:substitution | substitution:rule, substitution:cue, substitution:trap, substitution:app, substitution:step-1, substitution:step-2

ch04:lesson:redox | redox:rule, redox:cue, redox:trap, redox:app, redox:step-1, redox:step-2, redox:step-3

ch04:lesson:selectivity | selectivity:rule, selectivity:cue, selectivity:trap, selectivity:app, selectivity:step-1, selectivity:step-2

ch04:molecule:aceticacid | aceticacid

ch04:molecule:acetate | acetate

ch04:molecule:bromoethane | bromoethane

ch04:molecule:tbutylbromide | tbutylbromide

ch04:molecule:ethanol | ethanol

ch04:molecule:acetaldehyde | acetaldehyde

ch04:molecule:acetone | acetone

ch04:molecule:ketoester | ketoester

ch04:curriculum | scope-1, scope-2, scope-3, scope-4, scope-5

Curriculum scope

scope-1: 4.1 Acids and Bases

scope-2: 4.2 Nucleophiles, Electrophiles, and Leaving Groups

scope-3: 4.3 Oxidation–Reduction Reactions

scope-4: 4.4 Chemoselectivity

scope-5: 4.5 Steps to Problem Solving

Objective: Estimate acid–base equilibria

Objective: Distinguish SN1 and SN2

Objective: Track oxidation and chemoselectivity

Supplied Kaplan curriculum: Chapter 4; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter04b.xhtml

SN1 stereochemical qualifications: OpenStax 11.4 | https://openstax.org/books/organic-chemistry/pages/11-4-the-sn1-reaction

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

acids | Acid–base reasoning

Equilibrium favors the weaker acid/base pair

acids:rule | Core Rules

Brønsted acids donate H+ and bases accept it. Lewis acids accept an electron pair; Lewis bases donate one. For HA + B ⇌ A− + BH+, K ≈ 10^[pKa(BH+) − pKa(HA)] when the pKa values use comparable conditions.

acids:cue | Memory Cues

Lower pKa means stronger acid. Stabilize the conjugate base to increase acidity.

acids:trap | Common Mistakes / MCAT Traps

Basicity is an equilibrium property; nucleophilicity is a reaction-rate property. Do not use a single periodic trend across different solvents or unrelated functional groups.

acids:app | MCAT Applications

A carboxylic acid with pKa 4.8 is mostly carboxylate at pH 7.4; that changes water solubility and membrane partitioning.

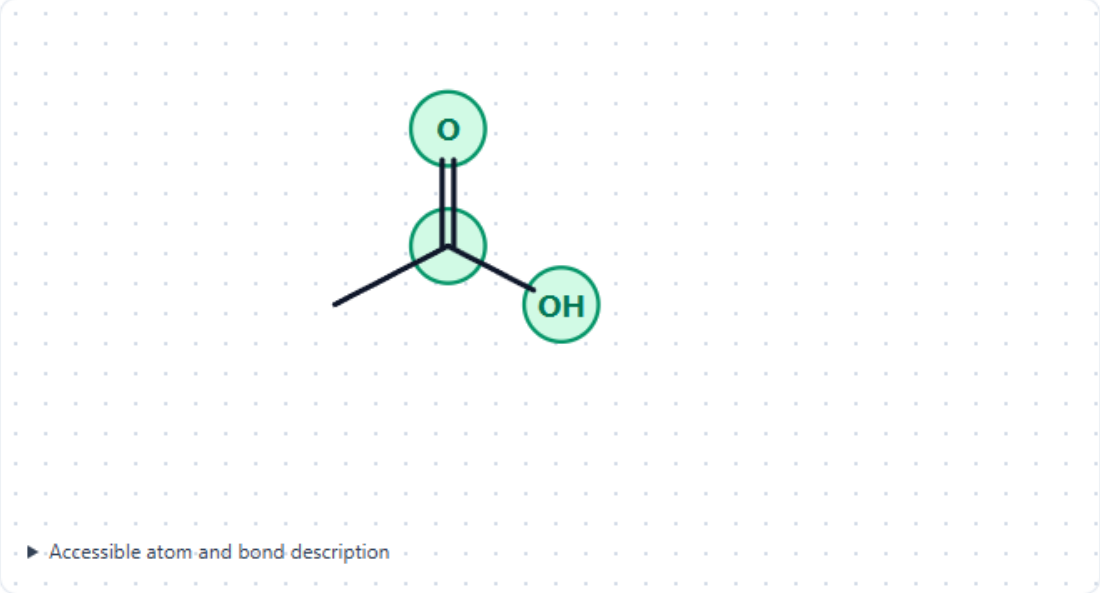
acids:step-1 | 1. Locate the acidic group

Removing the carboxyl O–H proton gives a carboxylate whose negative charge is shared across two oxygens.

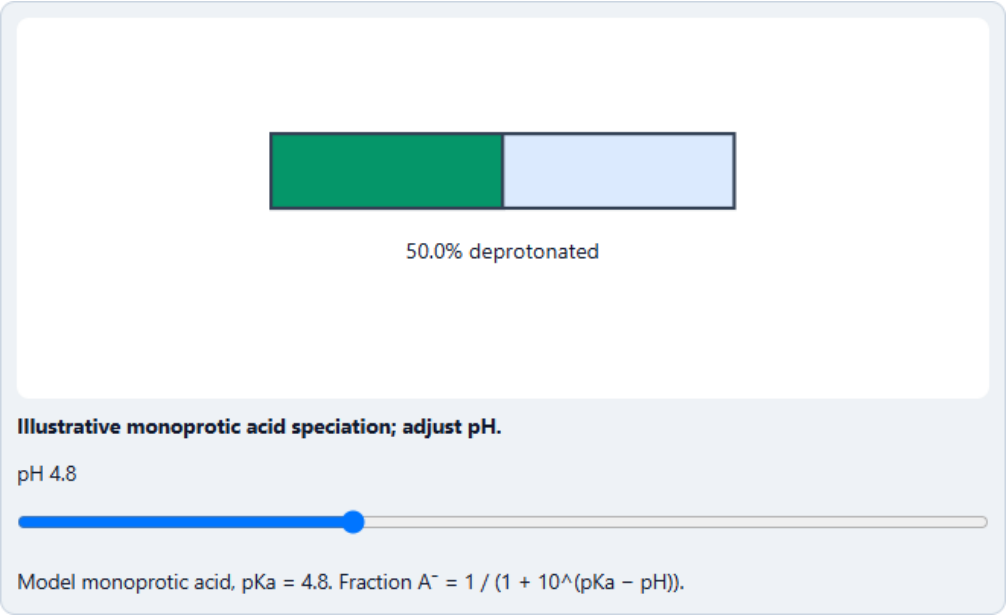
Displayed molecule ID: aceticacid

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: acids-step-1



Screenshot: acids-step-1-diagram



{"diagramSpecification": {"type": "acid", "pKa": 4.8, "caption": "Illustrative monoprotic acid speciation; adjust pH."}}

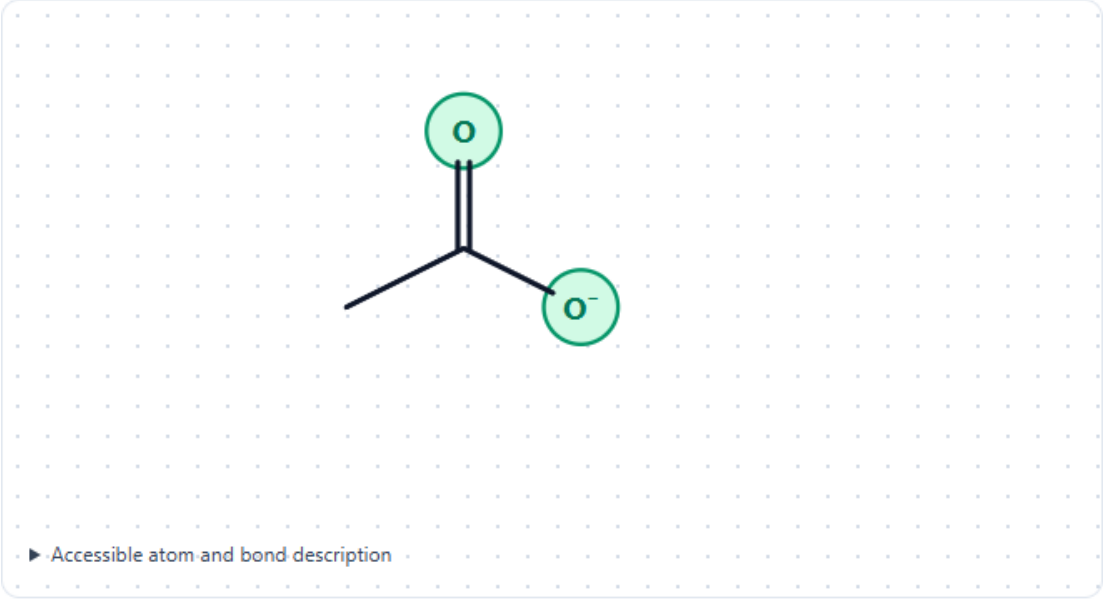
acids:step-2 | 2. Compare conjugate bases

An alcohol-derived alkoxide lacks the equivalent two-oxygen resonance of a carboxylate. This is a major reason simple carboxylic acids are more acidic than alcohols.

Displayed molecule ID: acetate

{"highlights": {"prin": [2, 3]}, "revealName": true}

Screenshot: acids-step-2



partners | Nucleophiles, electrophiles, and leaving groups

Electron pairs move from donor to acceptor

partners:rule | Core Rules

A nucleophile supplies an electron pair to an electrophile. A leaving group takes the bonding pair when it departs. Stable weak bases usually make better leaving groups. Protonation can turn poor leaving group OH− into neutral water.

partners:cue | Memory Cues

Find electron-rich donor → electron-poor acceptor → possible leaving group. Curved arrows start at electrons, not positive charges.

partners:trap | Common Mistakes / MCAT Traps

A carbonyl oxygen is electron rich while its carbon is electron poor. A positive charge is not required for electrophilicity. Polar protic solvents strongly solvate small anions; polar aprotic solvents often increase their SN2 reactivity.

partners:app | MCAT Applications

Enzyme active sites position nucleophiles and electrophiles while stabilizing charge; the same donor–acceptor logic applies.

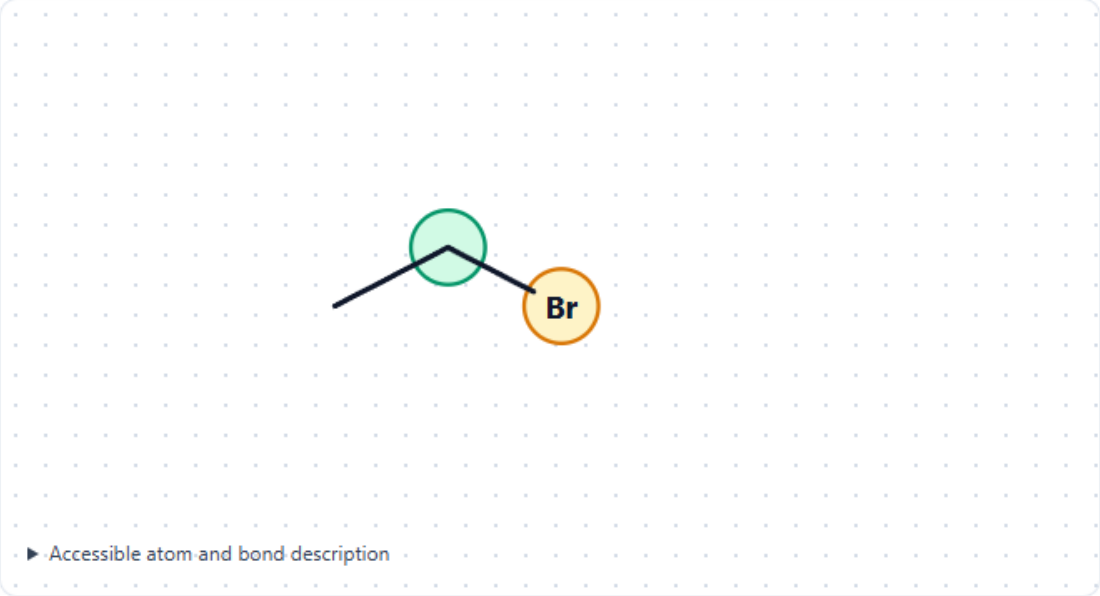
partners:step-1 | 1. Polarize the bond

Bromine withdraws electron density from the attached carbon. The C bonded to Br is the electrophilic site for substitution.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1], "sub": [2]}, "revealName": false}

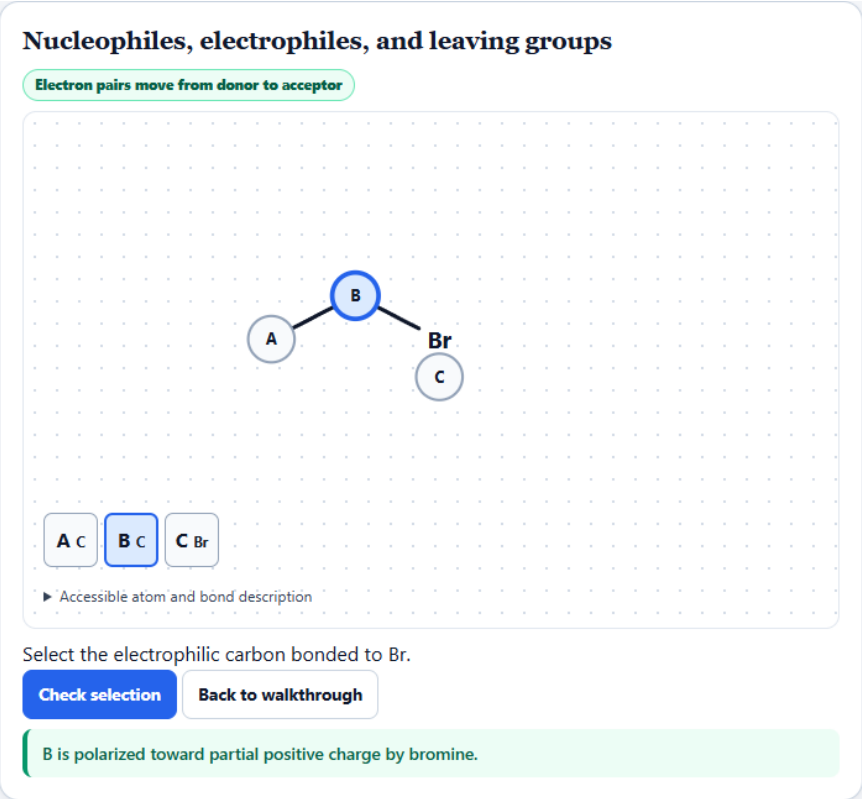
Screenshot: partners-step-1



Embedded challenge: Select the electrophilic carbon bonded to Br.

{"prompt": "Select the electrophilic carbon bonded to Br.", "answer": [1], "explanation": "B is polarized toward partial positive charge by bromine."}

Screenshot: partners-step-1-challenge



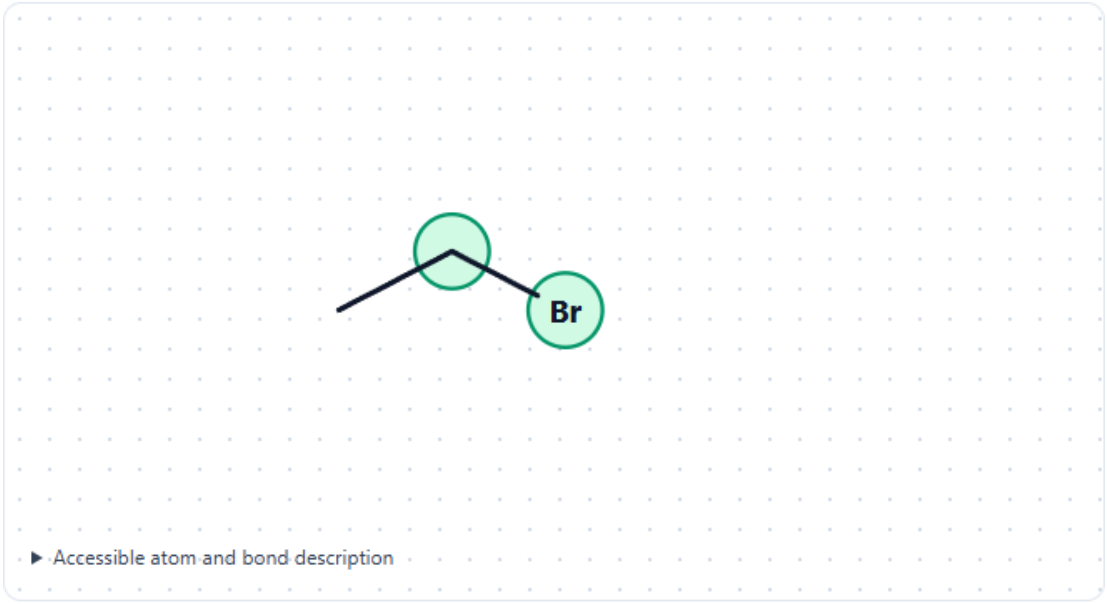
partners:step-2 | 2. Follow bond cleavage

In substitution, a nucleophile forms a bond to the electrophilic carbon while the C–Br bond ultimately breaks with its electron pair going to Br. Conditions determine whether this is concerted or stepwise.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: partners-step-2



substitution | SN1 vs. SN2

Mechanism depends on substrate and conditions

substitution:rule | Core Rules

SN2 is concerted backside attack: rate = k[substrate][nucleophile], with geometric inversion. Methyl and unhindered primary centers are favorable. SN1 ionizes to a carbocation: rate = k[substrate]; tertiary, allylic, and benzylic stabilization can help.

substitution:cue | Memory Cues

SN2: two species in the rate law, one concerted event. SN1: slow ionization, then attack.

substitution:trap | Common Mistakes / MCAT Traps

SN1 can rearrange and often racemizes a stereocenter, but ion pairing can prevent exactly 50:50 products. SN2 inversion does not guarantee an R↔S label change if CIP priorities change. Strong bases and heat can make elimination compete.

substitution:app | MCAT Applications

Use a rate change to distinguish mechanisms: doubling nucleophile concentration doubles an ideal SN2 rate but does not alter an ideal SN1 rate.

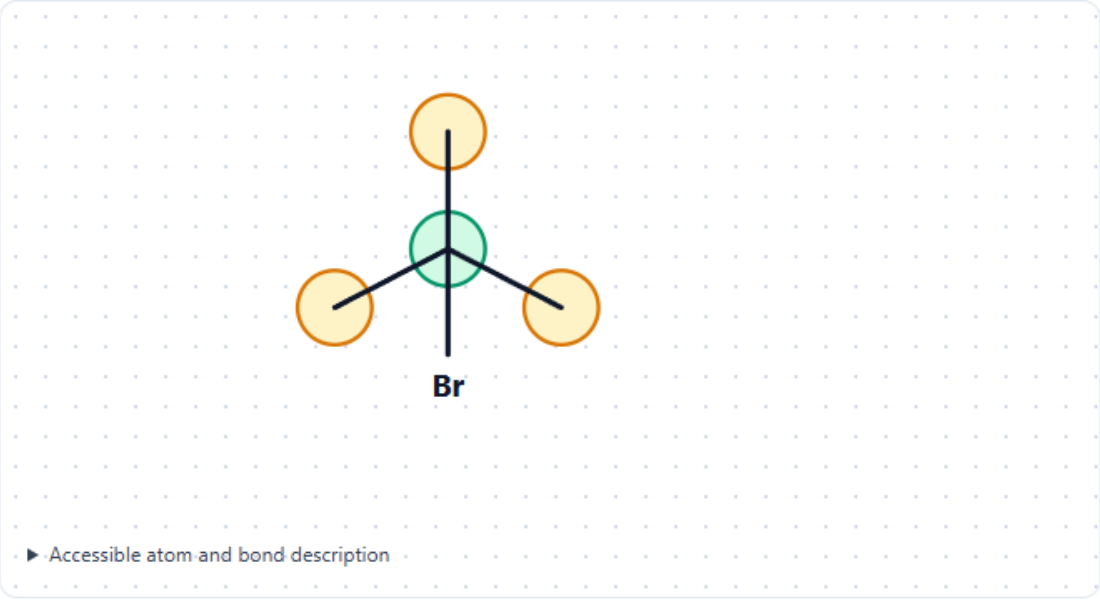
substitution:step-1 | 1. Inspect crowding

The carbon bonded to Br is tertiary, making backside SN2 attack strongly hindered. In a suitable ionizing solvent, ionization can lead to a tertiary carbocation.

Displayed molecule ID: tbutylbromide

{"highlights": {"prin": [1], "sub": [0, 2, 3]}, "revealName": true}

Screenshot: substitution-step-1



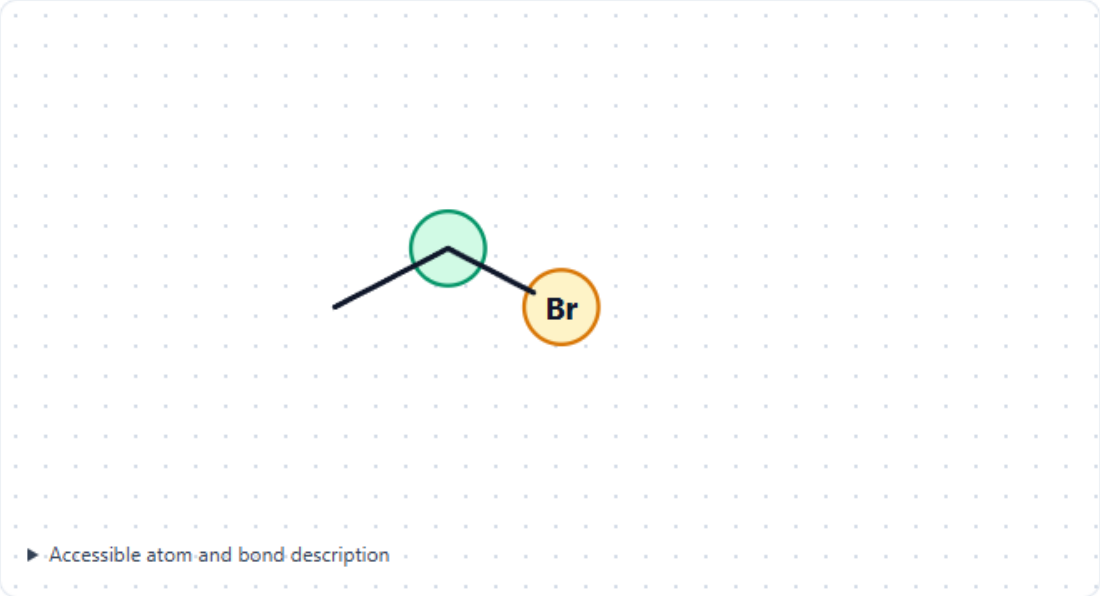
substitution:step-2 | 2. Compare a primary site

A primary carbon is less crowded. With a strong nucleophile in a suitable polar aprotic solvent, bromoethane is a much better SN2 candidate. A primary unstabilized carbocation is unfavorable.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1], "sub": [2]}, "revealName": true}

Screenshot: substitution-step-2



redox | Organic oxidation and reduction

Track bonds to H and electronegative atoms

redox:rule | Core Rules

Oxidation of carbon usually increases bonds to O/N/halogens or decreases C–H bonds. Reduction does the reverse. C–C bonds contribute zero when assigning carbon oxidation states. PCC under anhydrous conditions can oxidize a primary alcohol to an aldehyde; aqueous strong oxidants can give an acid.

redox:cue | Memory Cues

More bonds to O, less H: carbon is oxidized. Hydride addition to C=O generally reduces it.

redox:trap | Common Mistakes / MCAT Traps

NaBH4 commonly reduces aldehydes and ketones but ordinarily not esters or carboxylic acids. LiAlH4 reduces a wider set, including esters and acids, with appropriate workup. Reagent conditions matter.

redox:app | MCAT Applications

NAD+/NADH couples biological alcohol/carbonyl oxidation states through hydride transfer; not every laboratory reagent has a direct biological analogue.

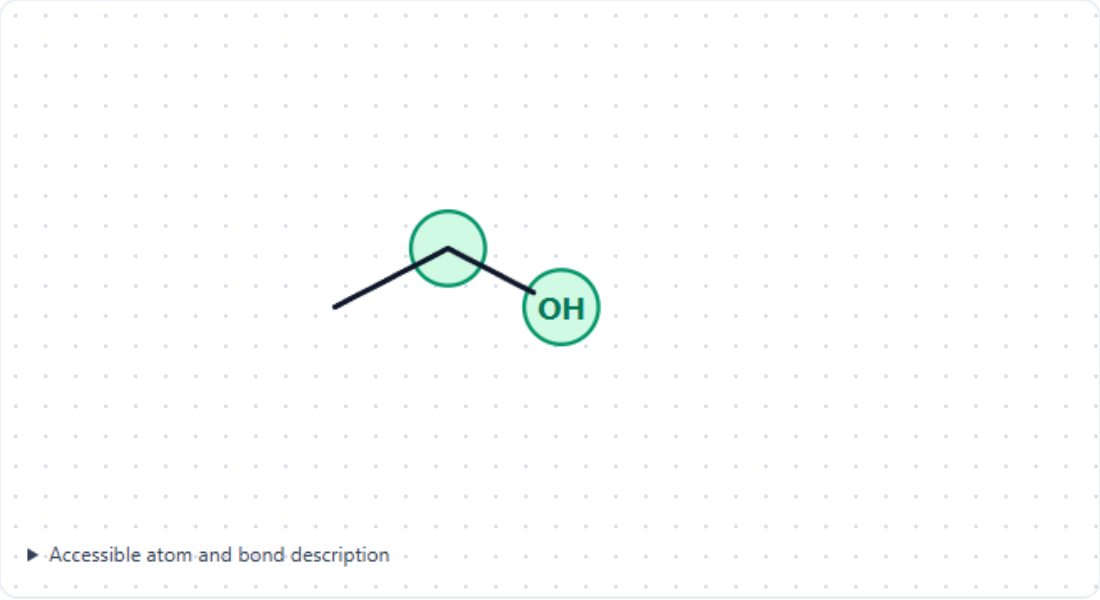
redox:step-1 | 1. Alcohol carbon

For the CH2OH carbon, one C–O bond contributes +1 and two C–H bonds contribute −2: oxidation state −1.

Displayed molecule ID: ethanol

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: redox-step-1



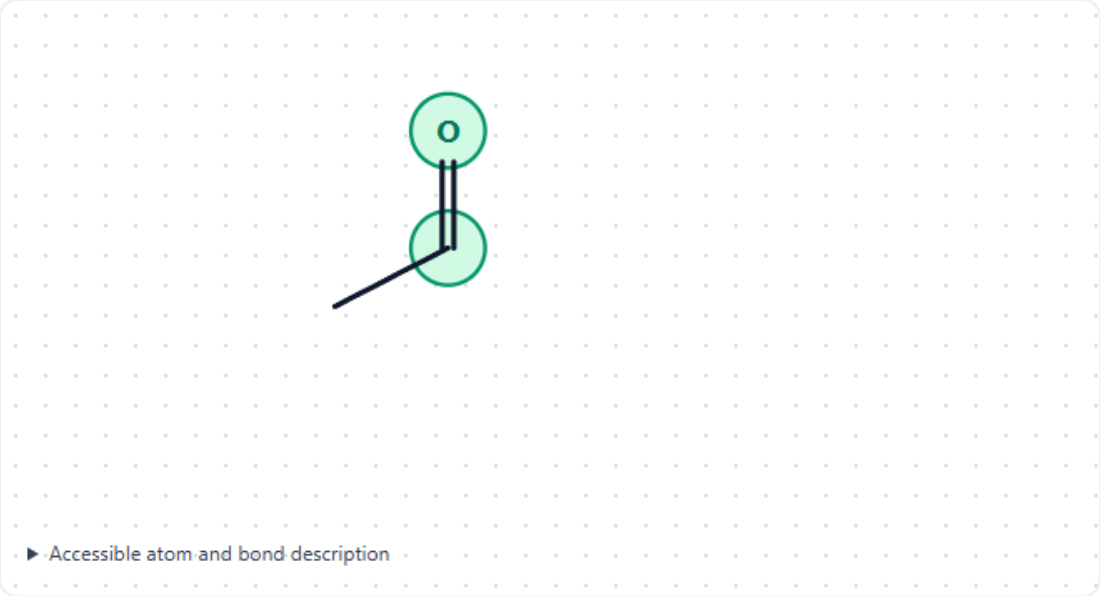
redox:step-2 | 2. Aldehyde carbon

For CHO, C=O contributes +2, C–H contributes −1, and C–C contributes 0: oxidation state +1. Alcohol → aldehyde is a two-electron oxidation.

Displayed molecule ID: acetaldehyde

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: redox-step-2



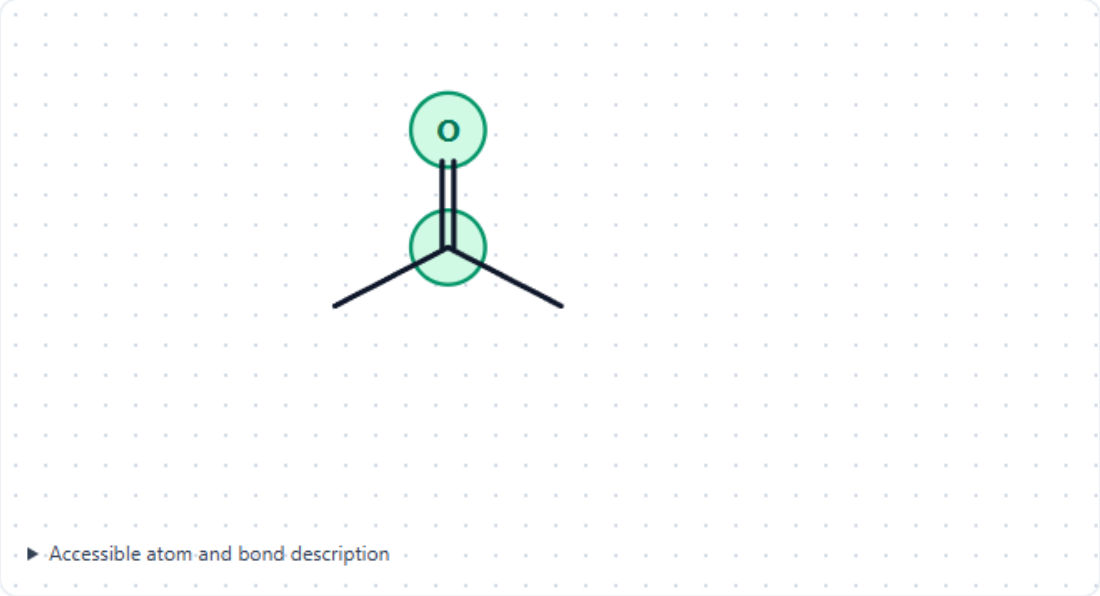
redox:step-3 | 3. Ketone reduction

Hydride adds to the electrophilic carbonyl carbon; protonation of oxygen gives a secondary alcohol.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: redox-step-3



selectivity | Chemoselectivity and a reaction workflow

Choose by reactivity, not naming seniority

selectivity:rule | Core Rules

Inventory functional groups, identify reagent roles, locate the likely reactive site, account for solvent and acid–base steps, then track bonds and stereochemistry. Chemoselectivity means preferential reaction at one functional group.

selectivity:cue | Memory Cues

Group inventory → reagent → first step → bond changes → product checks.

selectivity:trap | Common Mistakes / MCAT Traps

The group with the highest naming priority is not automatically the most reactive. Under ordinary NaBH4 conditions, a ketone can be reduced while an ester survives. Protecting groups change electronic reactivity, not merely steric bulk.

selectivity:app | MCAT Applications

Acetals protect aldehyde/ketone carbonyls from many basic and nucleophilic conditions; aqueous acid restores the carbonyl.

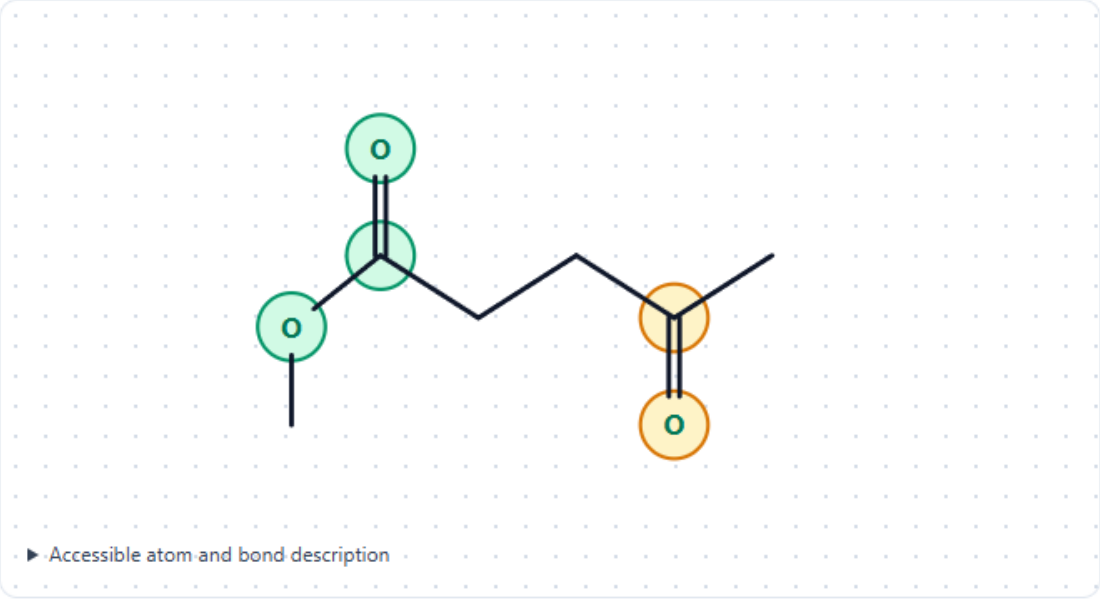
selectivity:step-1 | 1. Inventory two carbonyls

This molecule contains an ester at the left and a ketone at the right. The ester has an oxygen substituent directly on its carbonyl carbon.

Displayed molecule ID: ketoester

{"highlights": {"prin": [0, 5, 6], "sub": [3, 8]}, "revealName": true}

Screenshot: selectivity-step-1



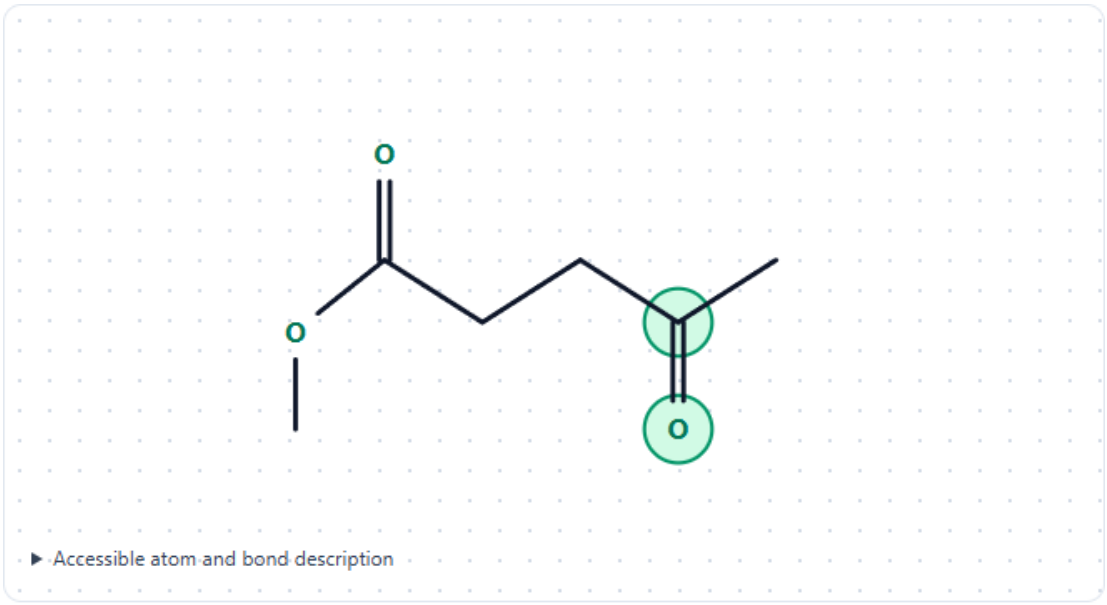
selectivity:step-2 | 2. Predict a selective step

Ordinary NaBH4 conditions favor reduction of the ketone. Ester resonance donation lowers its electrophilicity relative to the ketone; a stronger reducing system can change the outcome.

Displayed molecule ID: ketoester

{"highlights": {"prin": [3, 8]}, "revealName": false}

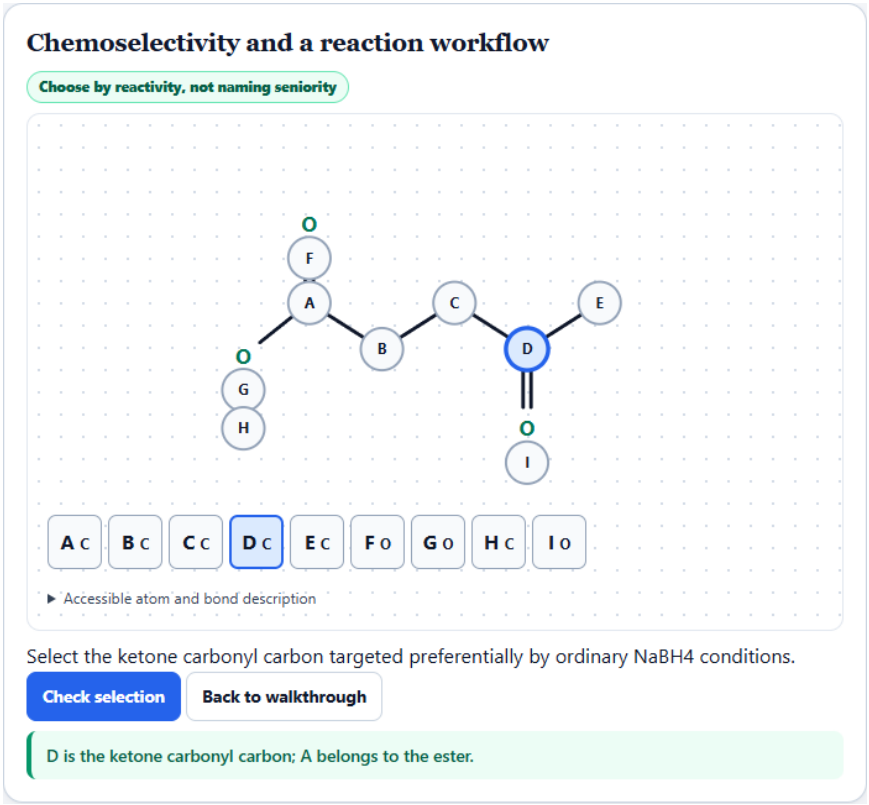
Screenshot: selectivity-step-2



Embedded challenge: Select the ketone carbonyl carbon targeted preferentially by ordinary NaBH4 conditions.

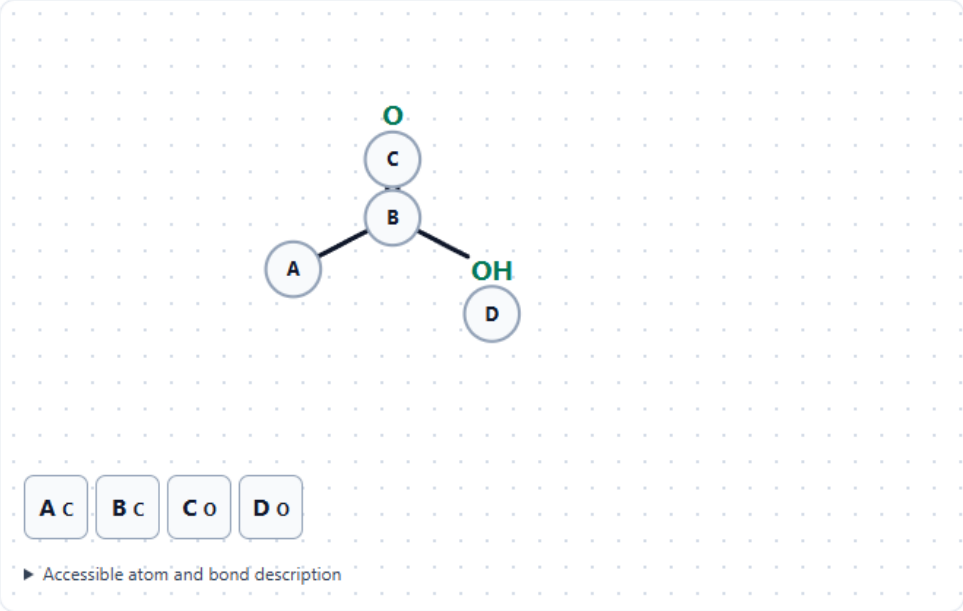
{"prompt": "Select the ketone carbonyl carbon targeted preferentially by ordinary NaBH4 conditions.", "answer": [3], "explanation": "D is the ketone carbonyl carbon; A belongs to the ester."}

Screenshot: selectivity-step-2-challenge



pka-equilibrium | Estimate acid–base equilibrium

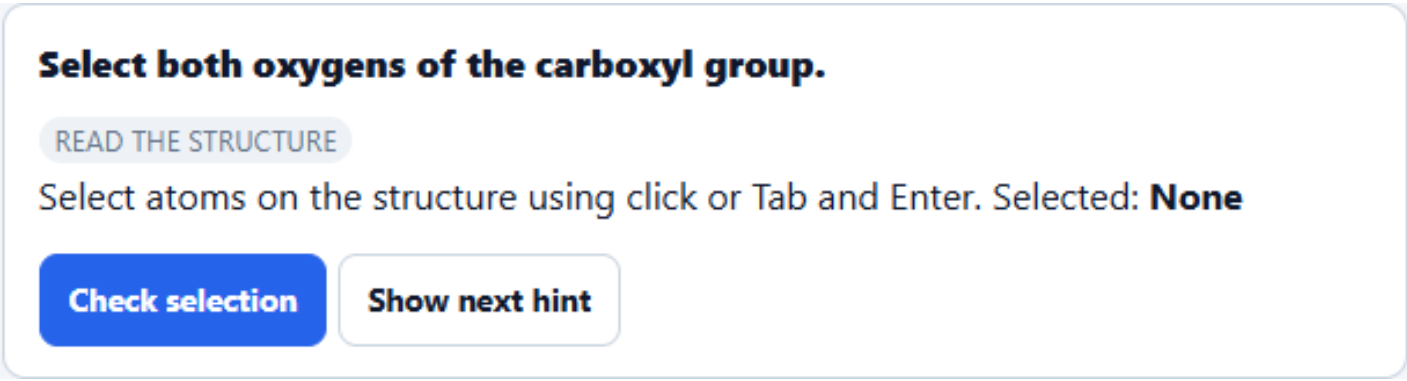
Screenshot: pka-equilibrium-stage-1-molecule



pka-equilibrium:stage-1 | Select both oxygens of the carboxyl group.

Answer type: atoms

Screenshot: pka-equilibrium-stage-1-question



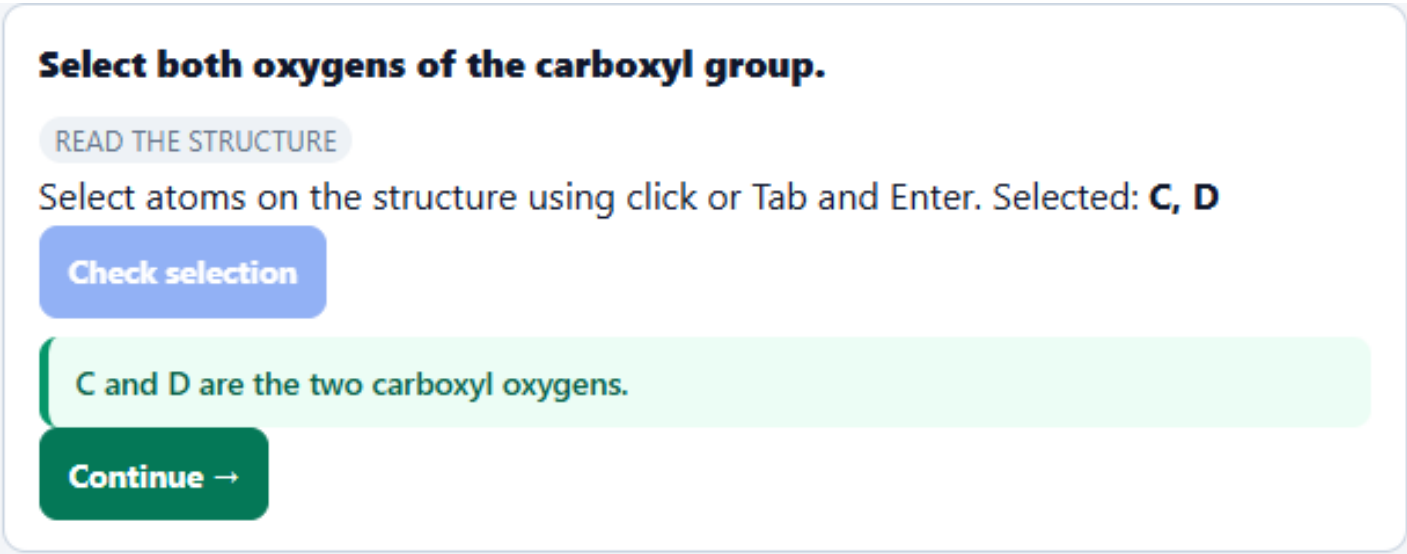
{"authorAnswer": [2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 3]}}

Hint 1: Locate C=O and O–H on the same carbon.

Hint 2: Select the two O labels.

Explanation: C and D are the two carboxyl oxygens.

Screenshot: pka-equilibrium-stage-1-feedback



pka-equilibrium:stage-2 | If pKa(HA)=5 and pKa(BH+)=10, equilibrium favors...

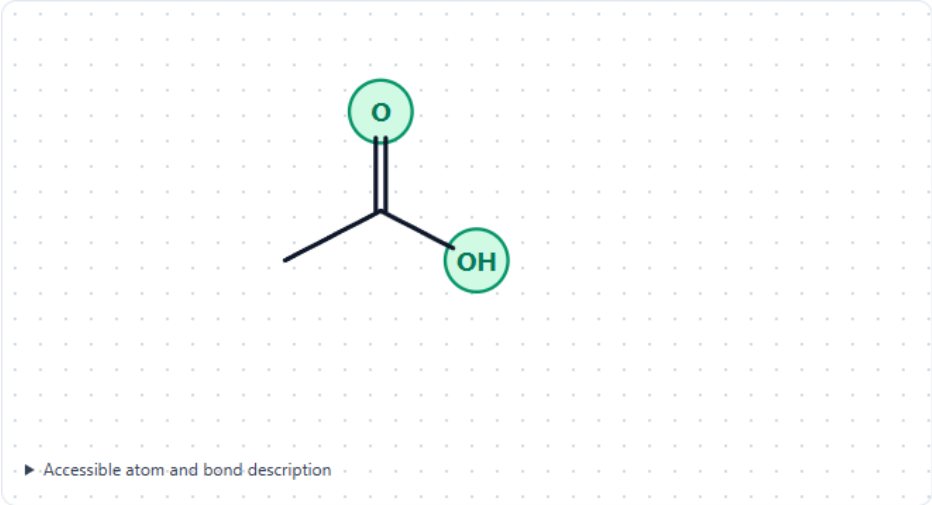
Answer type: choice

Option 0 (A): Reactants

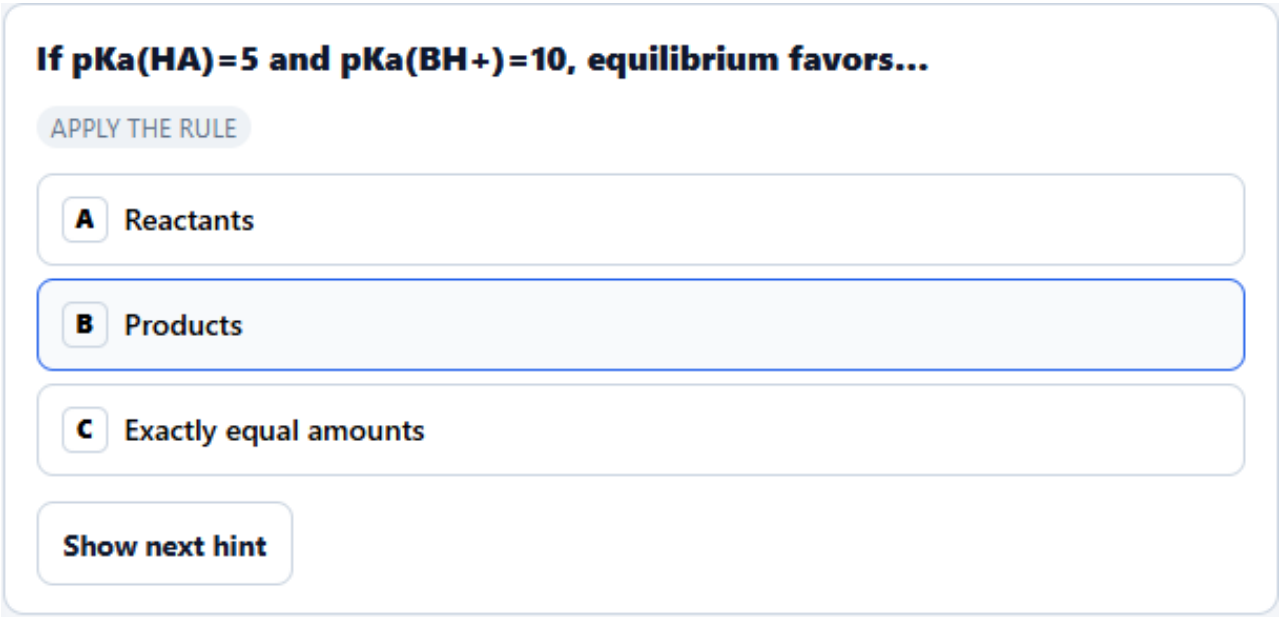
Option 1 (B): Products

Option 2 (C): Exactly equal amounts

Screenshot: pka-equilibrium-stage-2-molecule



Screenshot: pka-equilibrium-stage-2-question



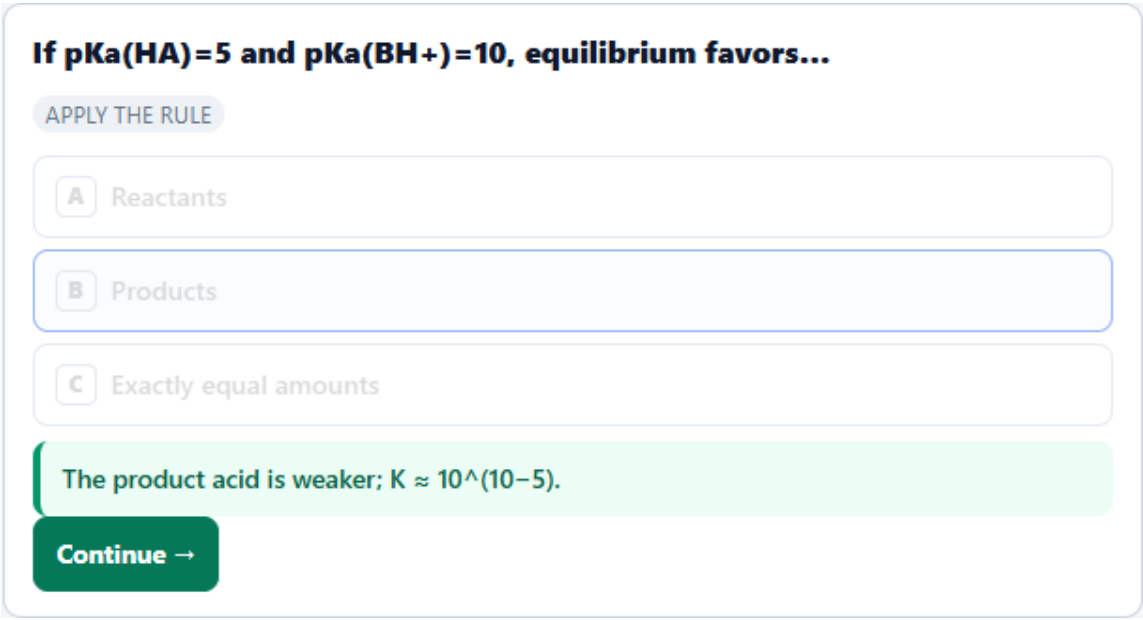
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare the two acids.

Hint 2: Equilibrium favors the side with the higher-pKa acid.

Explanation: The product acid is weaker; K ≈ 10^(10−5).

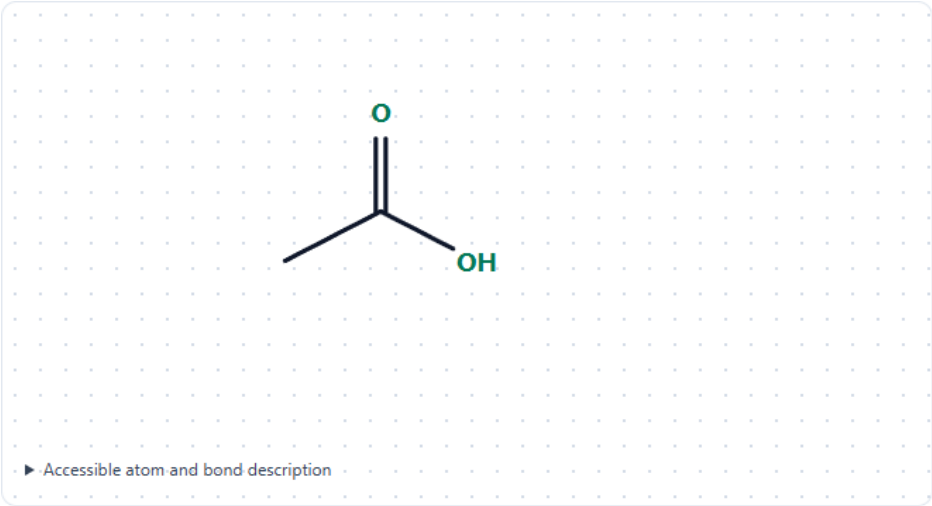
Screenshot: pka-equilibrium-stage-2-feedback



pka-equilibrium:stage-3 | Estimate K for those pKa values.

Answer type: number | unit: dimensionless

Screenshot: pka-equilibrium-stage-3-molecule



Screenshot: pka-equilibrium-stage-3-question



{"authorAnswer": 100000, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Subtract reactant-acid pKa from product-acid pKa.

Hint 2: Raise ten to that difference.

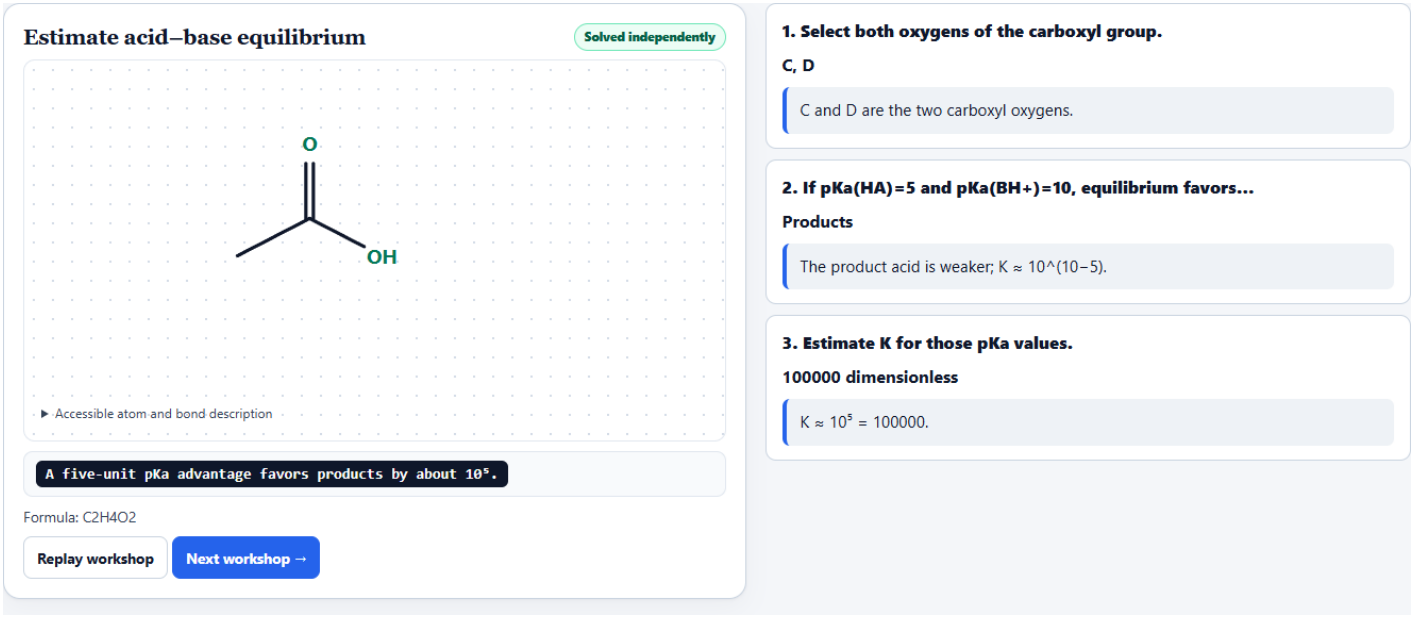
Explanation: $K \approx 10^5 = 100000$.

Screenshot: pka-equilibrium-stage-3-feedback



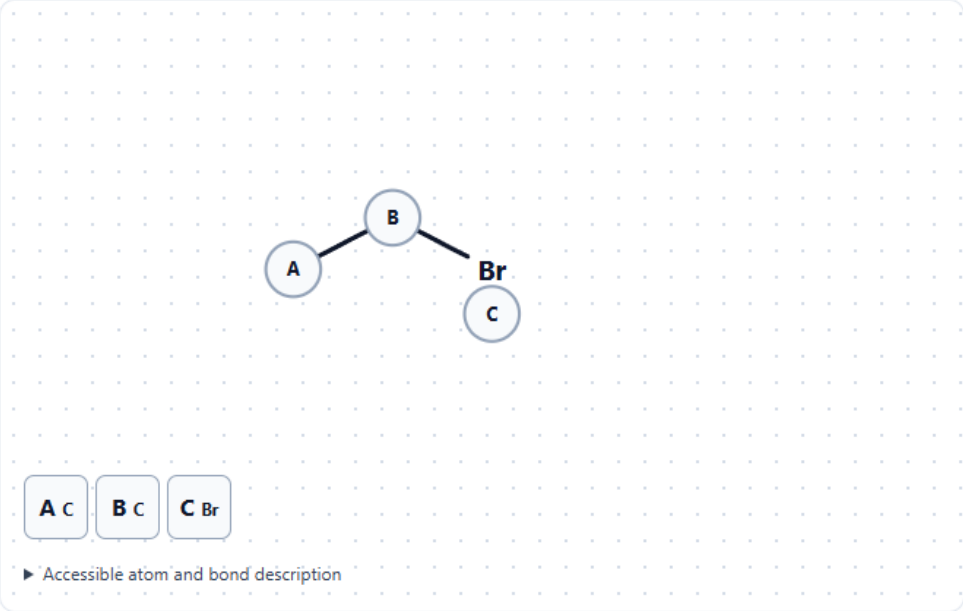
Completion summary: A five-unit pKa advantage favors products by about 10^5.

Screenshot: pka-equilibrium-complete



identify-partners | Find the substitution partners

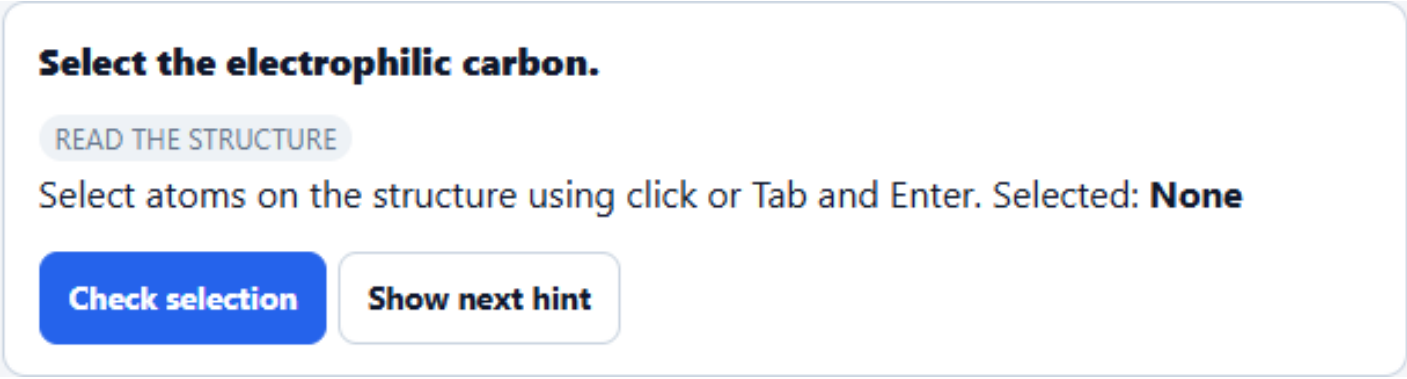
Screenshot: identify-partners-stage-1-molecule



identify-partners:stage-1 | Select the electrophilic carbon.

Answer type: atoms

Screenshot: identify-partners-stage-1-question



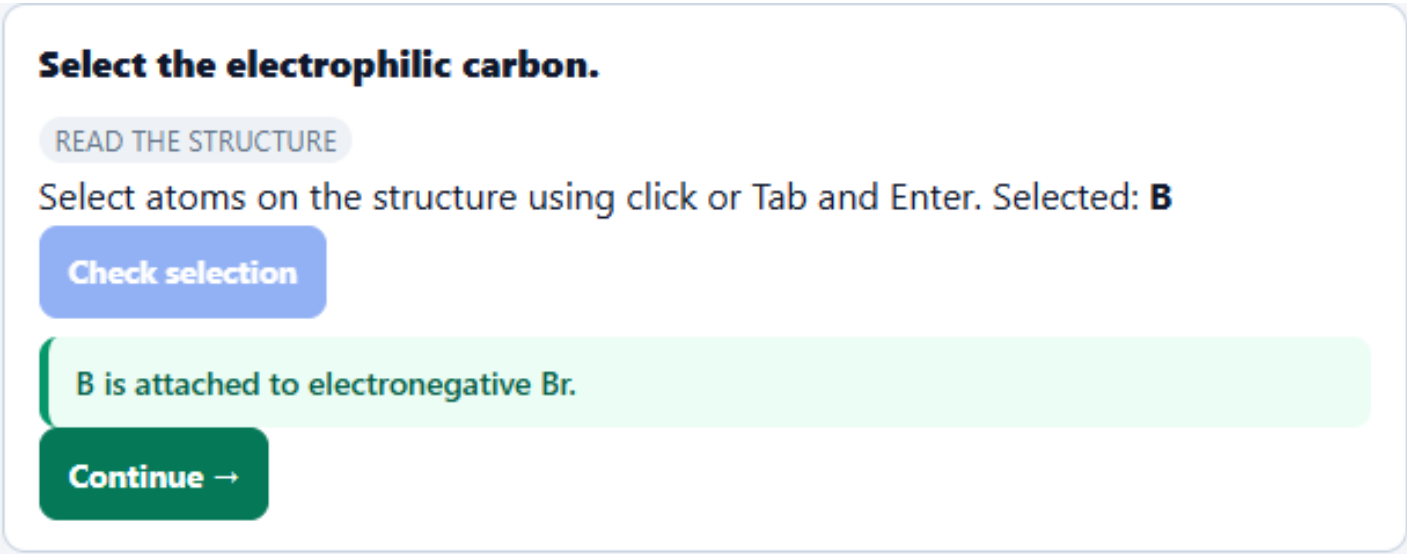
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1]}}

Hint 1: Look for the carbon bonded to the leaving group.

Hint 2: The polarized C–Br bond identifies it.

Explanation: B is attached to electronegative Br.

Screenshot: identify-partners-stage-1-feedback



identify-partners:stage-2 | Which species supplies a lone pair as a nucleophile?

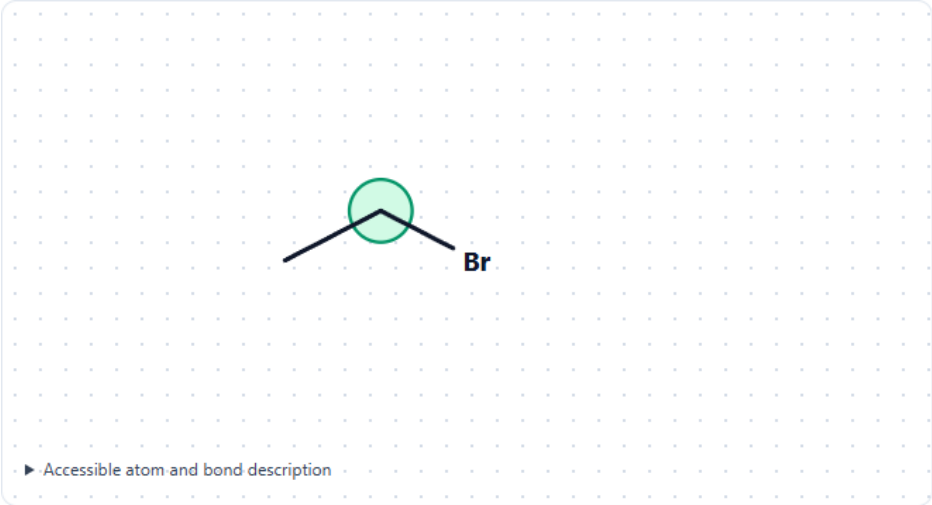
Answer type: choice

Option 0 (A): Hydroxide ion

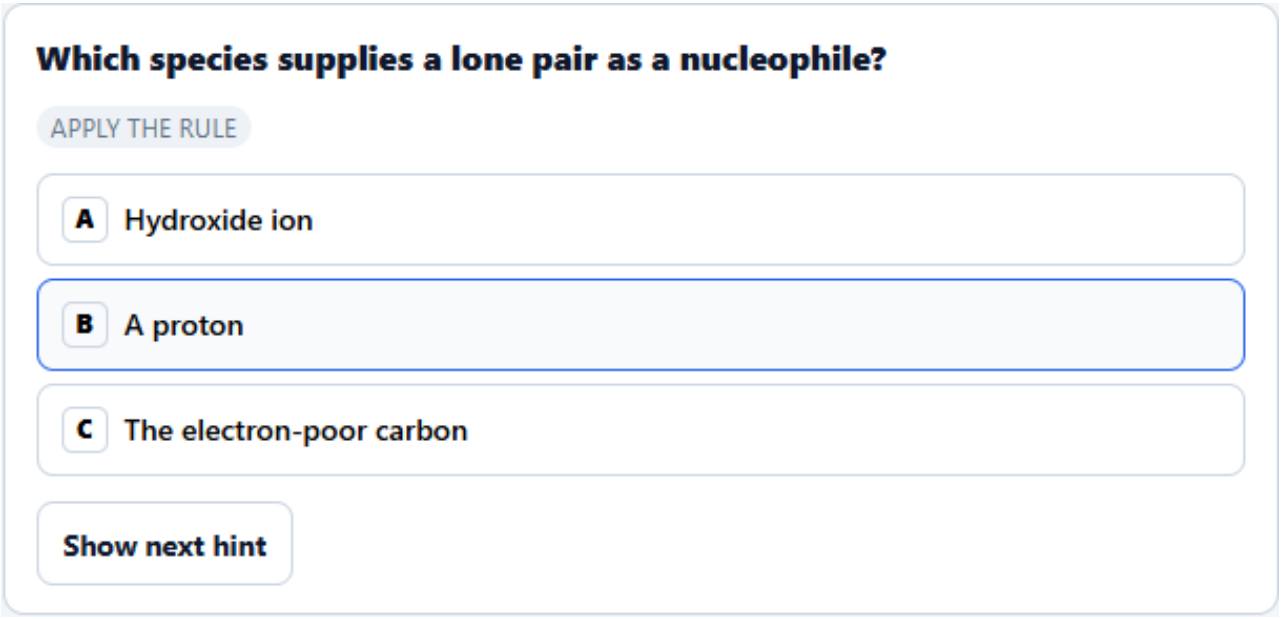
Option 1 (B): A proton

Option 2 (C): The electron-poor carbon

Screenshot: identify-partners-stage-2-molecule



Screenshot: identify-partners-stage-2-question



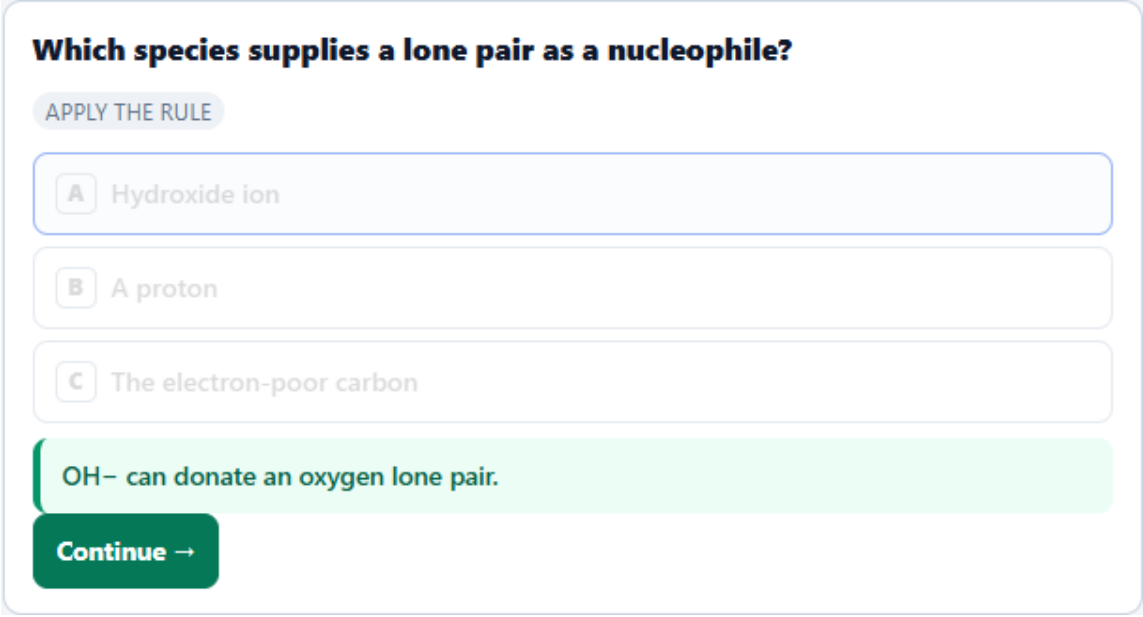
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Nucleophiles are electron-pair donors.

Hint 2: A negative charge often indicates available electrons.

Explanation: OH− can donate an oxygen lone pair.

Screenshot: identify-partners-stage-2-feedback



identify-partners:stage-3 | When C–Br breaks heterolytically, the pair goes to…

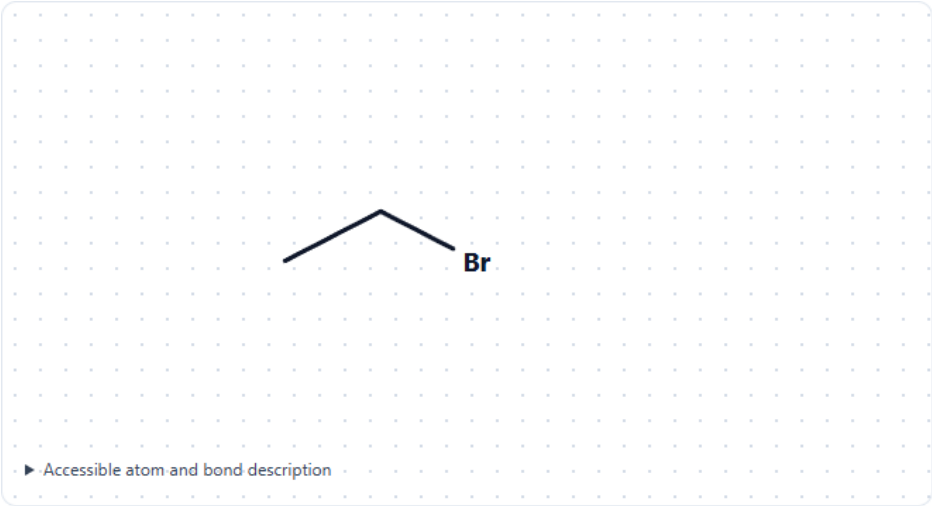
Answer type: choice

Option 0 (A): Carbon

Option 1 (B): Bromine

Option 2 (C): Neither atom

Screenshot: identify-partners-stage-3-molecule



Screenshot: identify-partners-stage-3-question



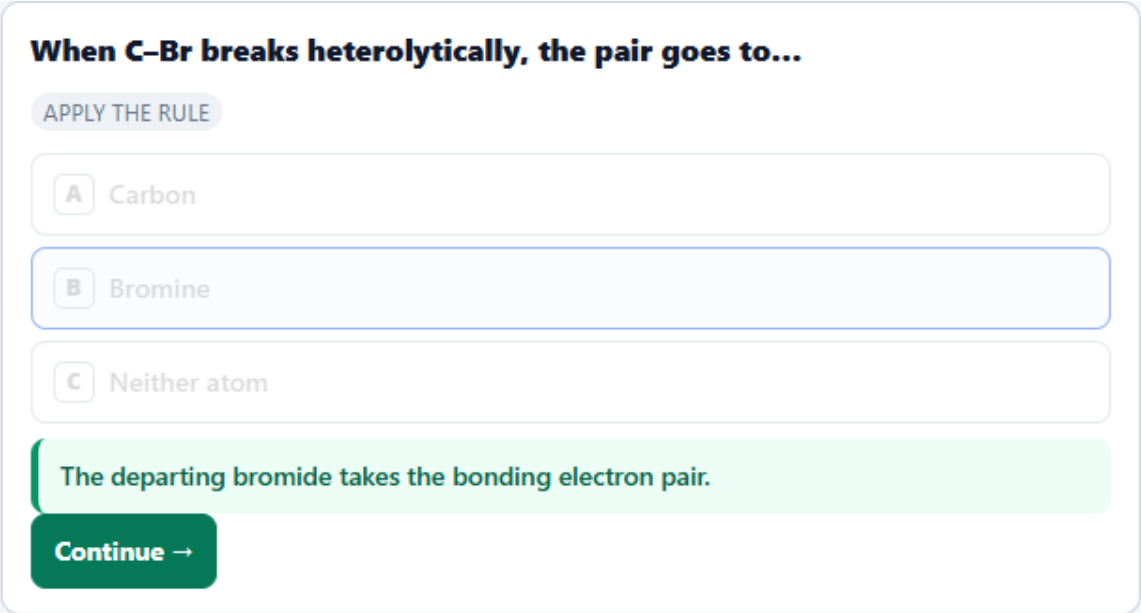
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: A leaving group retains the bond electrons.

Hint 2: Bromide is a relatively stable anion.

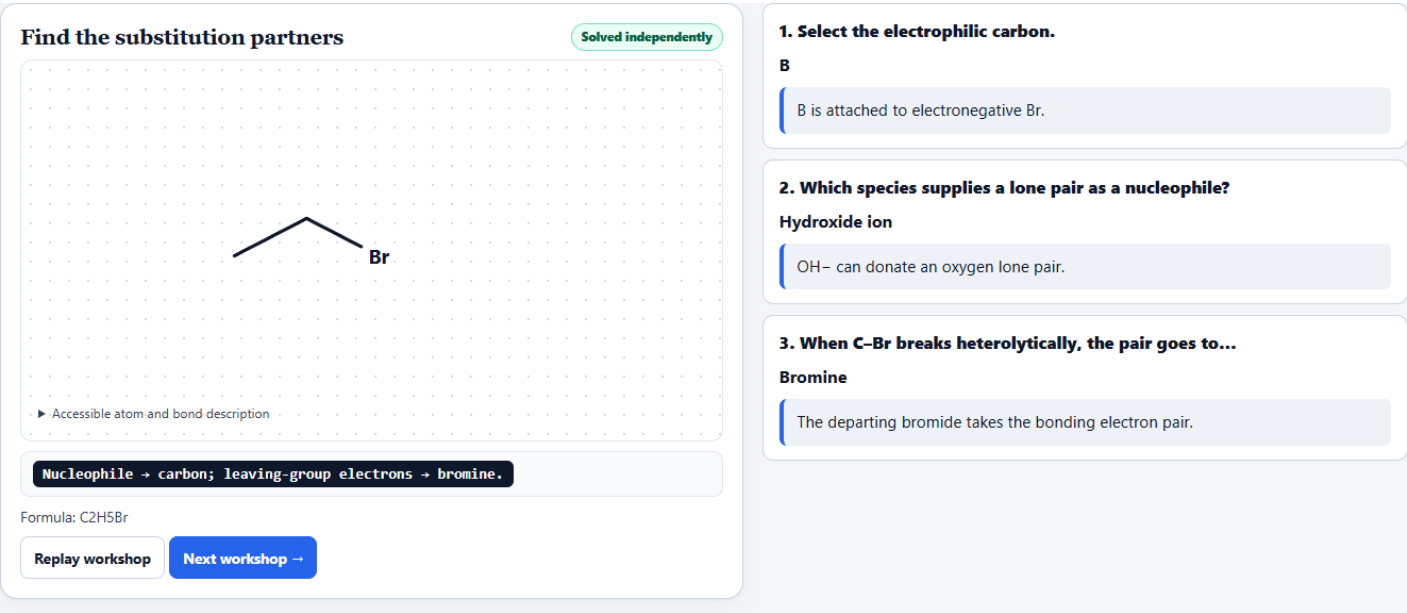
Explanation: The departing bromide takes the bonding electron pair.

Screenshot: identify-partners-stage-3-feedback



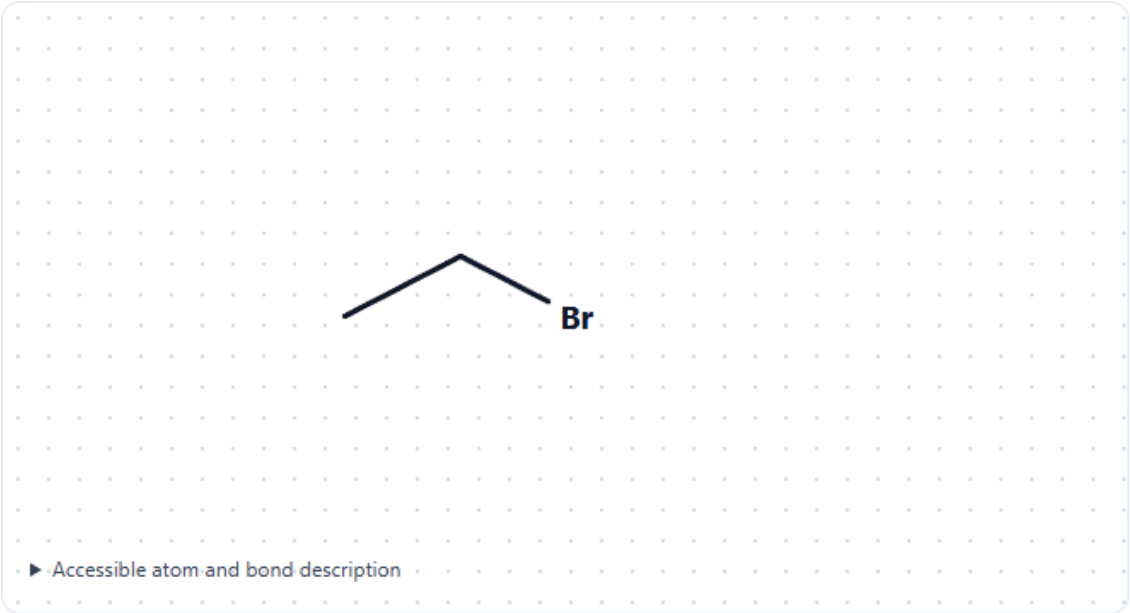
Completion summary: Nucleophile → carbon; leaving-group electrons → bromine.

Screenshot: identify-partners-complete



sn2-rate | Mechanism and kinetics

Screenshot: sn2-rate-stage-1-molecule



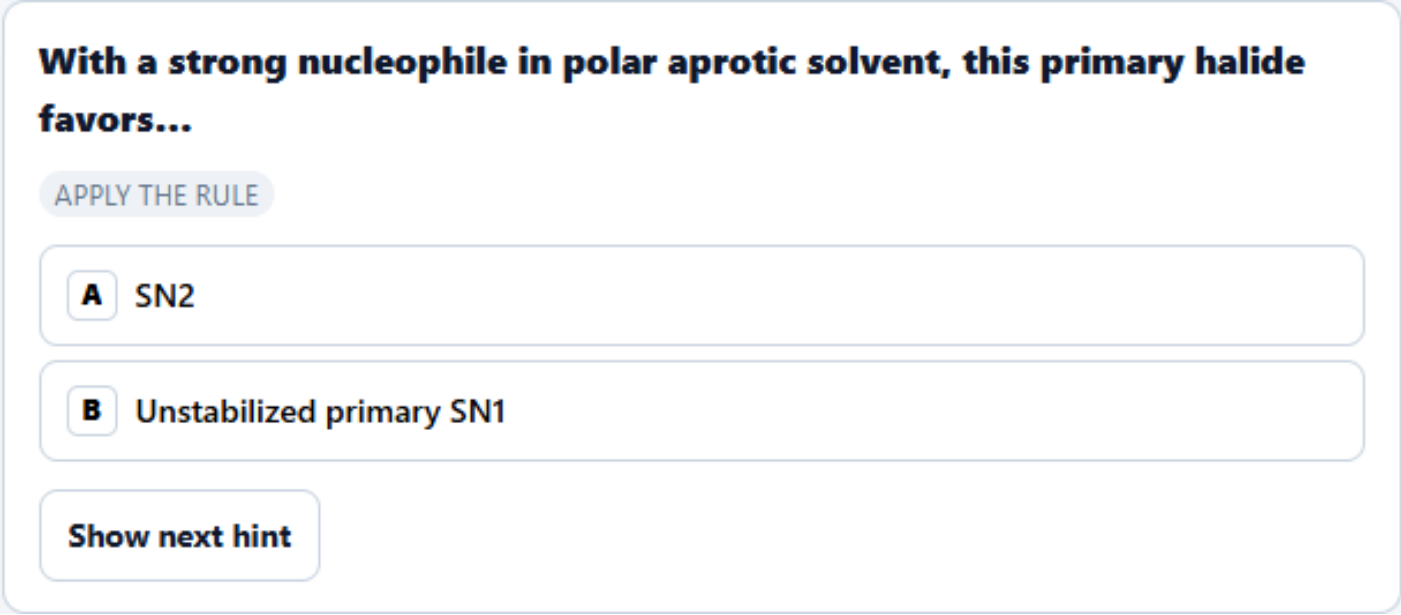
sn2-rate:stage-1 | With a strong nucleophile in polar aprotic solvent, this primary halide favors...

Answer type: choice

Option 0 (A): SN2

Option 1 (B): Unstabilized primary SN1

Screenshot: sn2-rate-stage-1-question



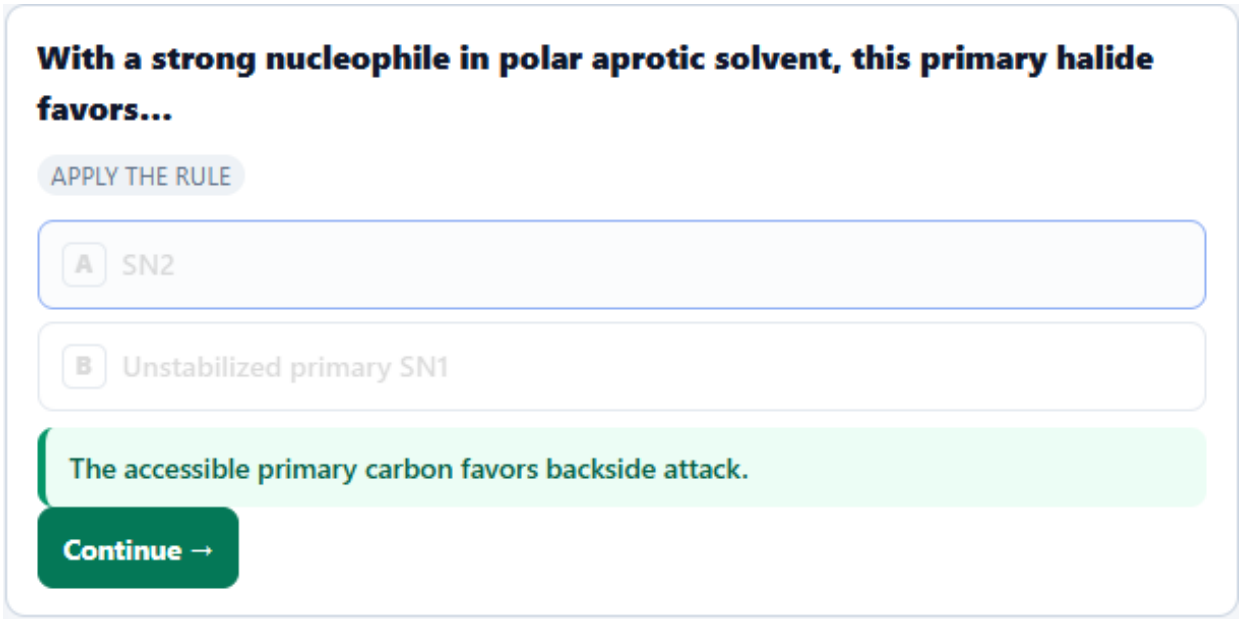
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Assess steric access and carbocation stability.

Hint 2: An unstabilized primary carbocation is poor.

Explanation: The accessible primary carbon favors backside attack.

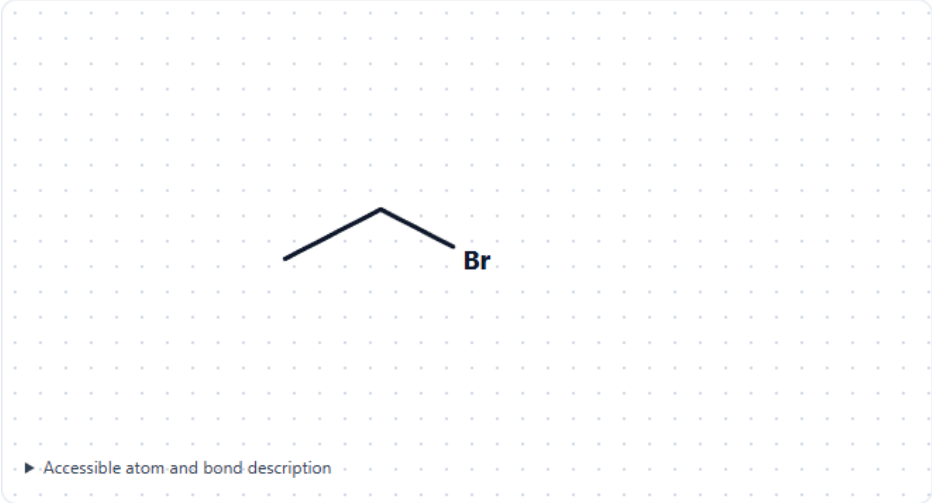
Screenshot: sn2-rate-stage-1-feedback



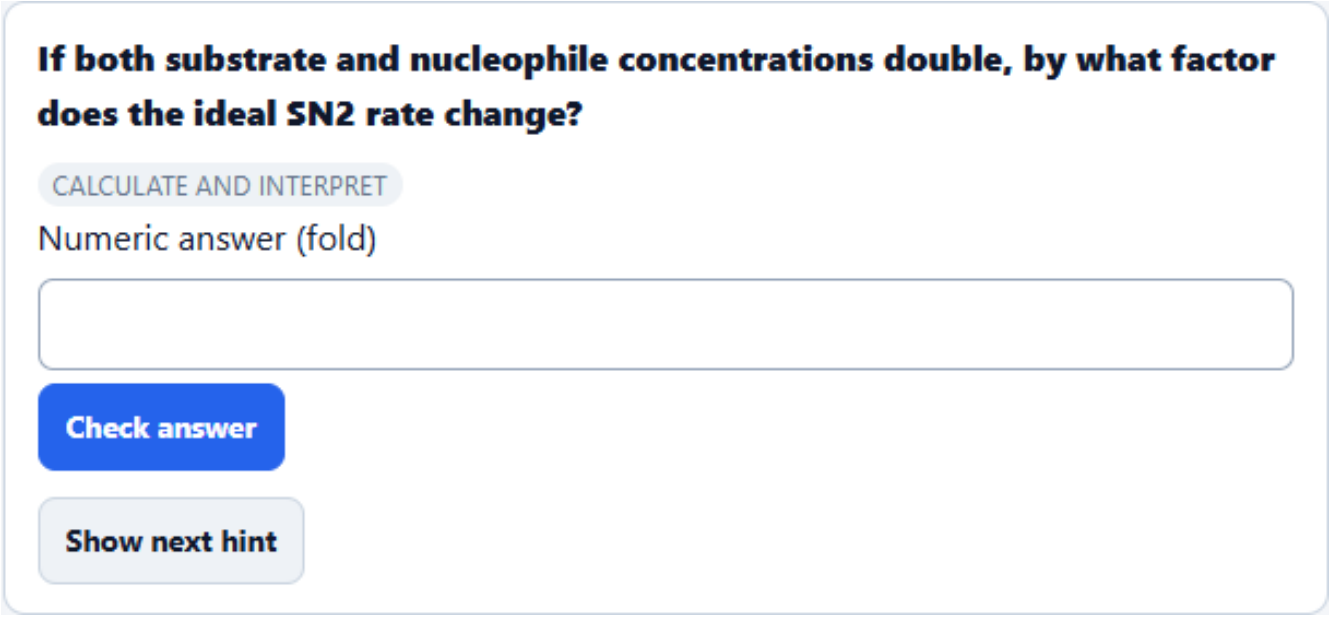
sn2-rate:stage-2 | If both substrate and nucleophile concentrations double, by what factor does the ideal SN2 rate change?

Answer type: number | unit: fold

Screenshot: sn2-rate-stage-2-molecule



Screenshot: sn2-rate-stage-2-question



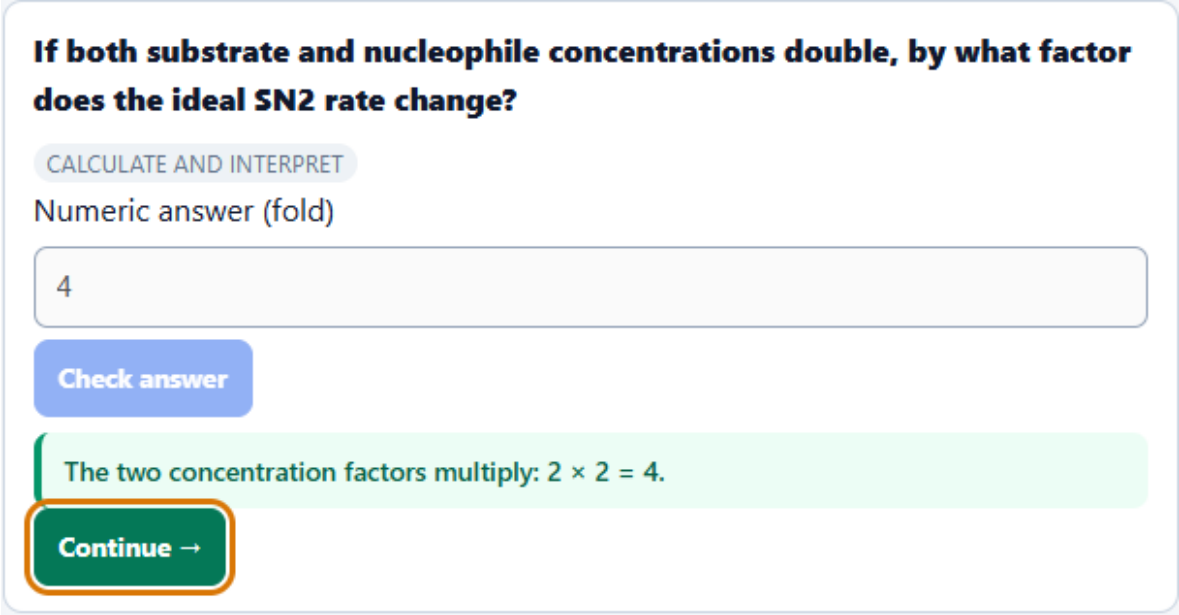
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use rate = k[substrate][nucleophile].

Hint 2: Each concentration contributes one factor of two.

Explanation: The two concentration factors multiply: 2 × 2 = 4.

Screenshot: sn2-rate-stage-2-feedback



sn2-rate:stage-3 | At a chiral SN2 reaction center, the geometry undergoes...

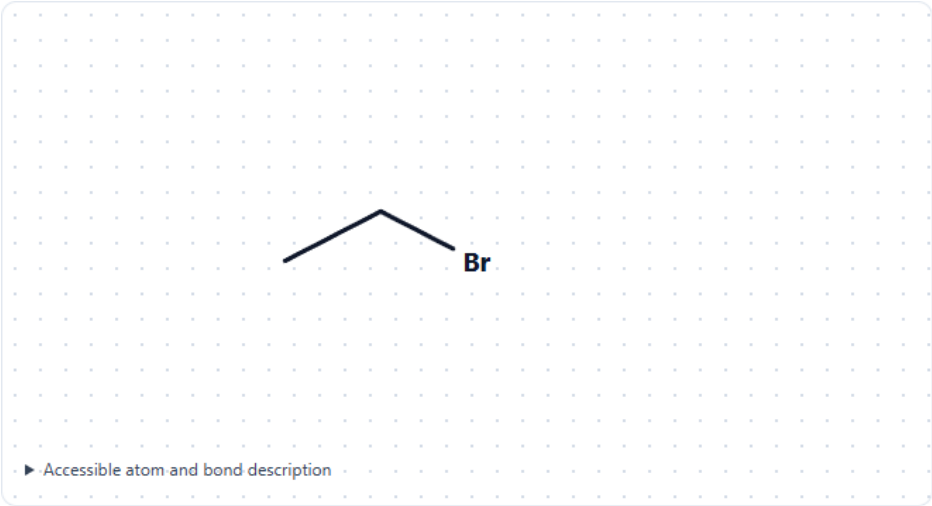
Answer type: choice

Option 0 (A): Inversion

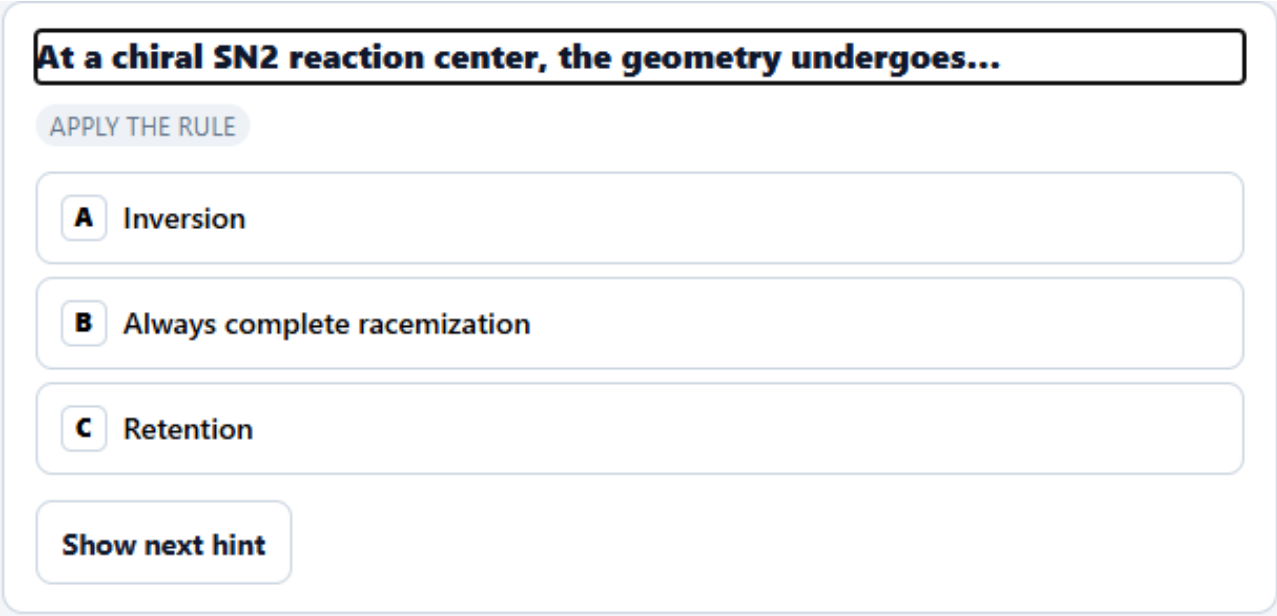
Option 1 (B): Always complete racemization

Option 2 (C): Retention

Screenshot: sn2-rate-stage-3-molecule



Screenshot: sn2-rate-stage-3-question



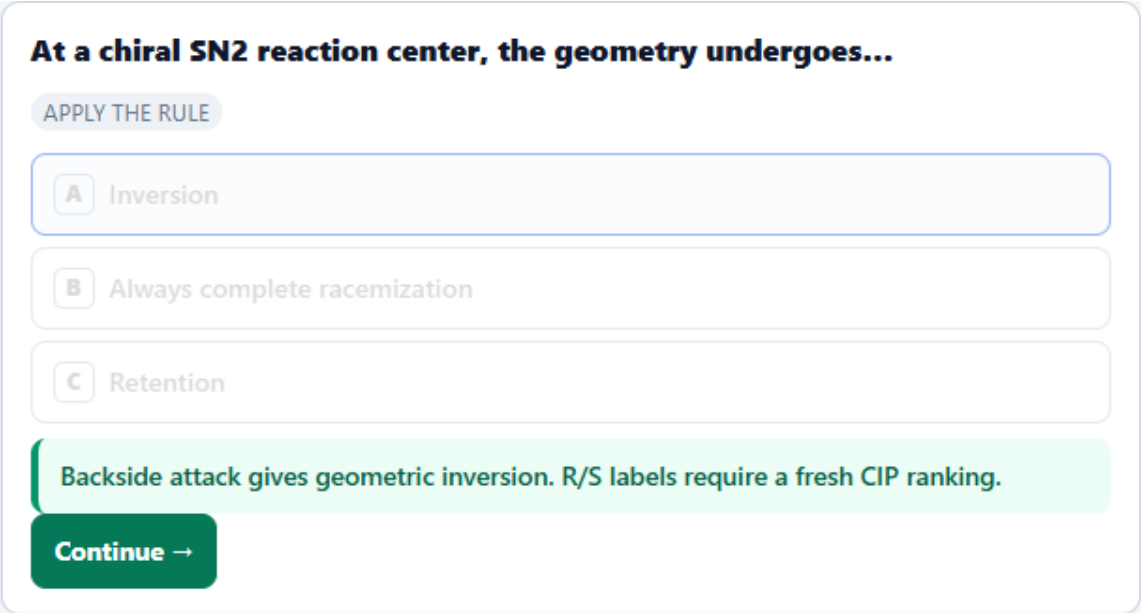
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The nucleophile approaches opposite the leaving group.

Hint 2: Think of an umbrella turning inside out.

Explanation: Backside attack gives geometric inversion. R/S labels require a fresh CIP ranking.

Screenshot: sn2-rate-stage-3-feedback



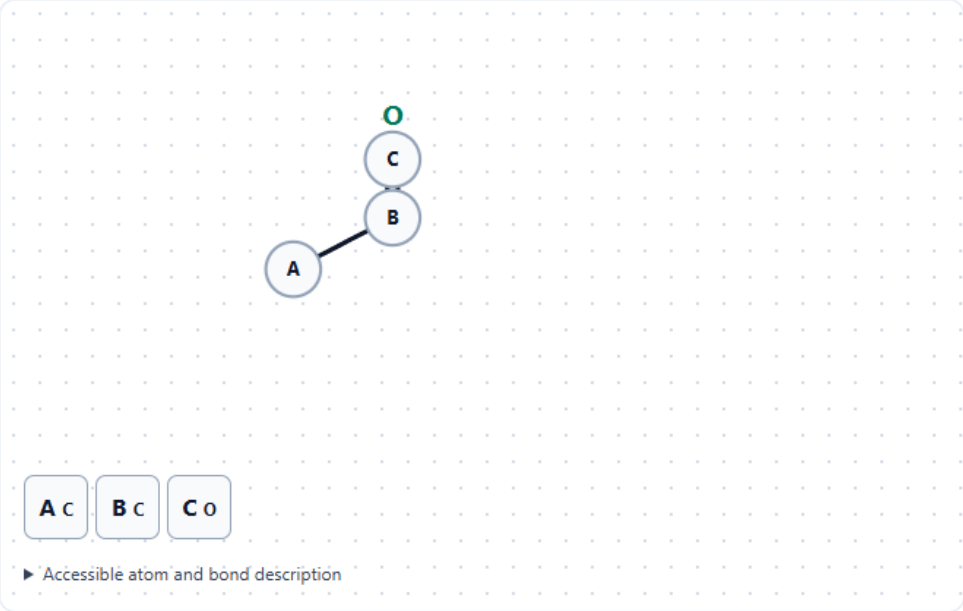
Completion summary: SN2 is bimolecular and stereospecific for geometric inversion.

Screenshot: sn2-rate-complete



carbon-redox | Assign carbon oxidation state

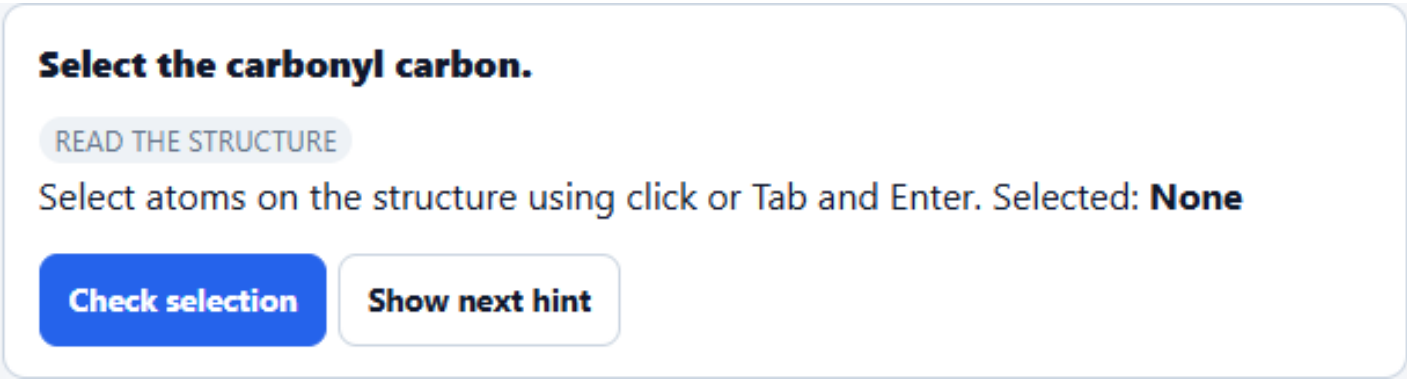
Screenshot: carbon-redox-stage-1-molecule



carbon-redox:stage-1 | Select the carbonyl carbon.

Answer type: atoms

Screenshot: carbon-redox-stage-1-question



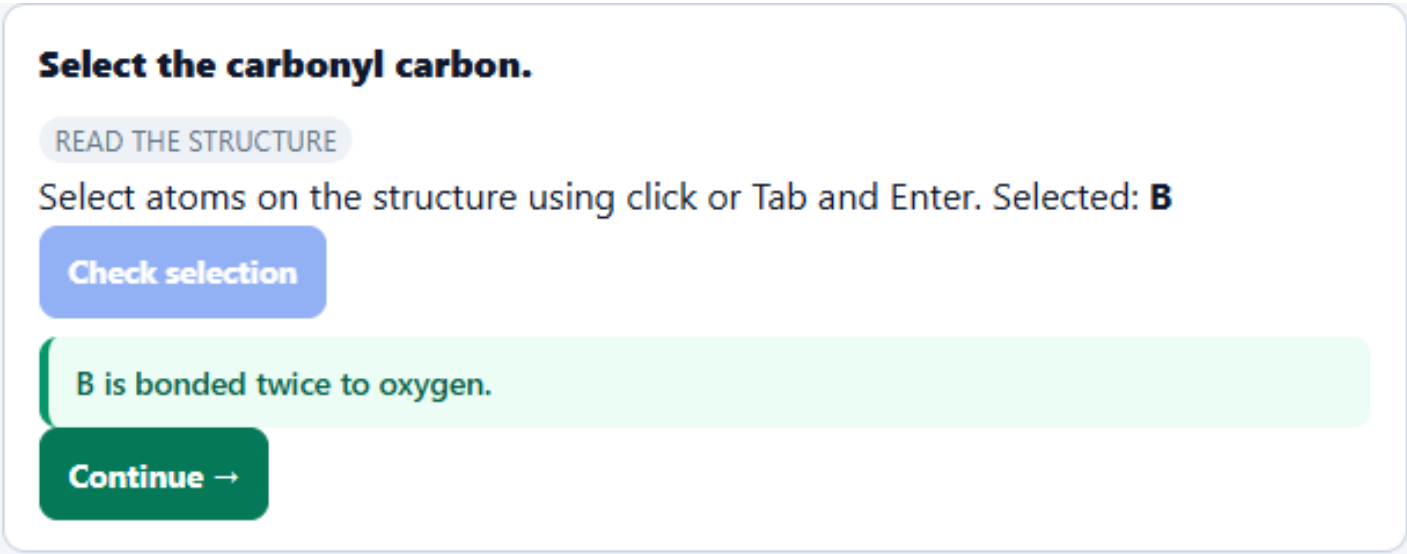
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Find the C=O bond.

Hint 2: Choose its carbon endpoint.

Explanation: B is bonded twice to oxygen.

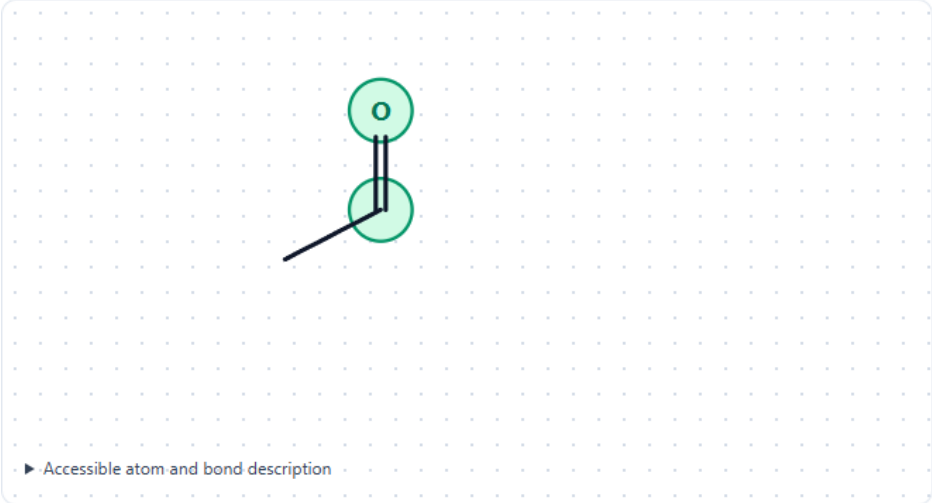
Screenshot: carbon-redox-stage-1-feedback



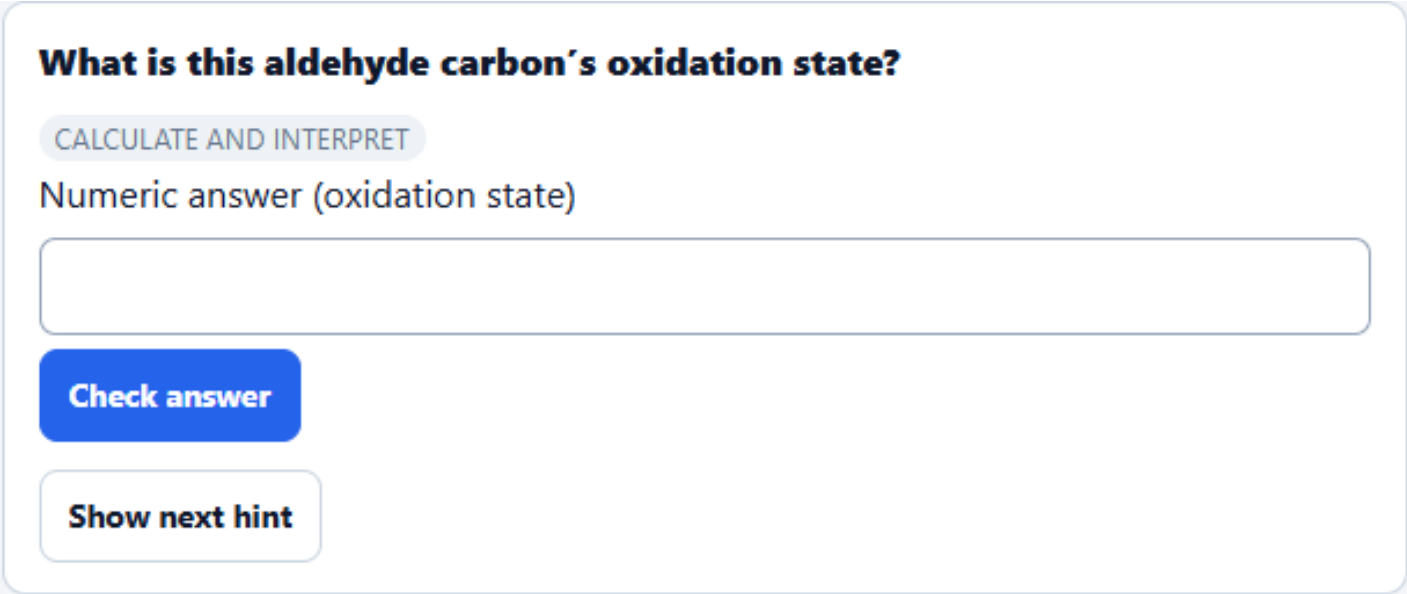
carbon-redox:stage-2 | What is this aldehyde carbon’s oxidation state?

Answer type: number | unit: oxidation state

Screenshot: carbon-redox-stage-2-molecule



Screenshot: carbon-redox-stage-2-question



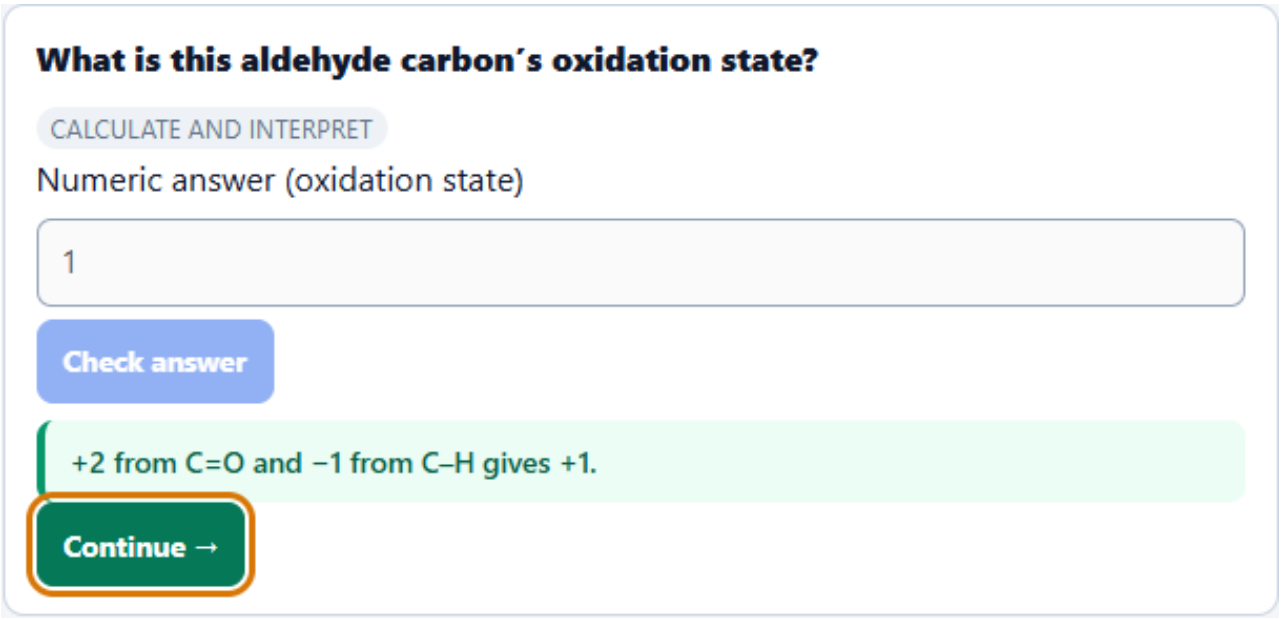
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Assign electrons in C–O to O and in C–H to C.

Hint 2: C–C contributes zero.

Explanation: +2 from C=O and −1 from C–H gives +1.

Screenshot: carbon-redox-stage-2-feedback



carbon-redox:stage-3 | NaBH4 followed by protonation converts it to…

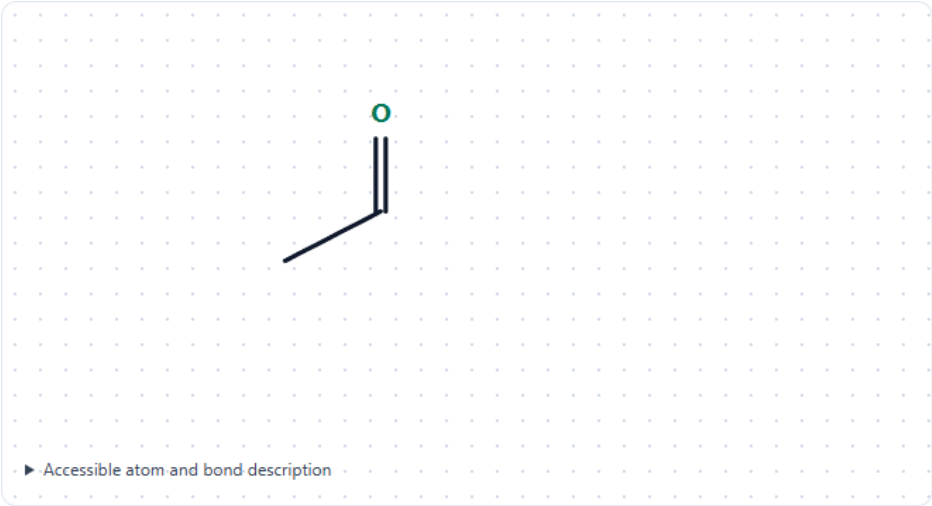
Answer type: choice

Option 0 (A): A primary alcohol

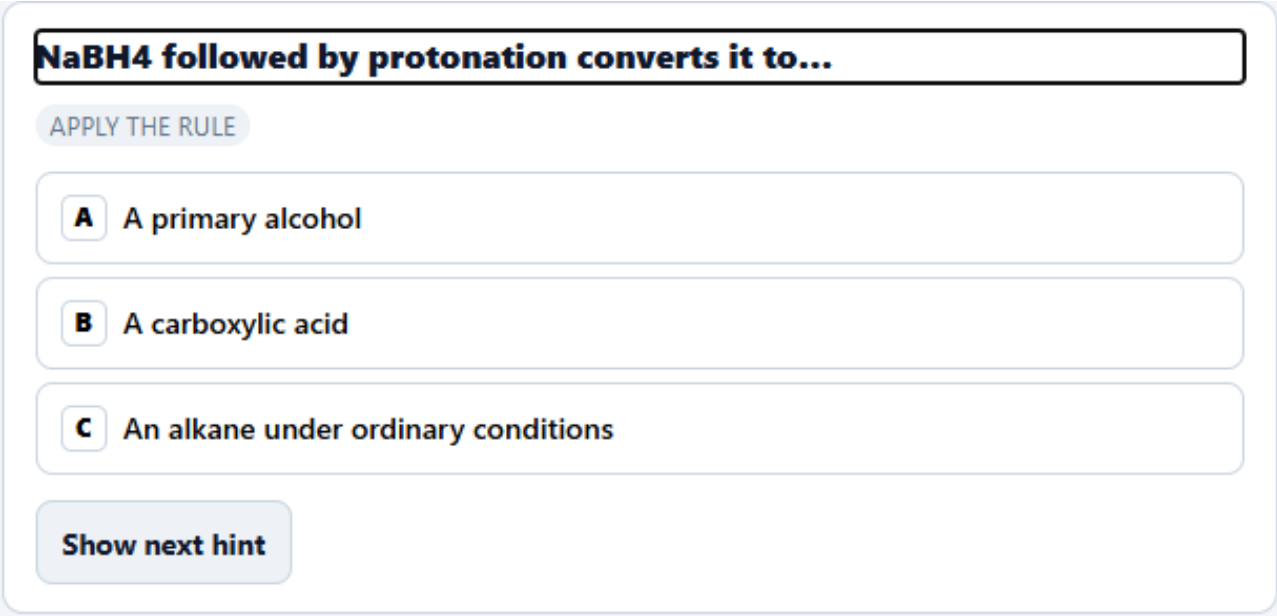
Option 1 (B): A carboxylic acid

Option 2 (C): An alkane under ordinary conditions

Screenshot: carbon-redox-stage-3-molecule



Screenshot: carbon-redox-stage-3-question



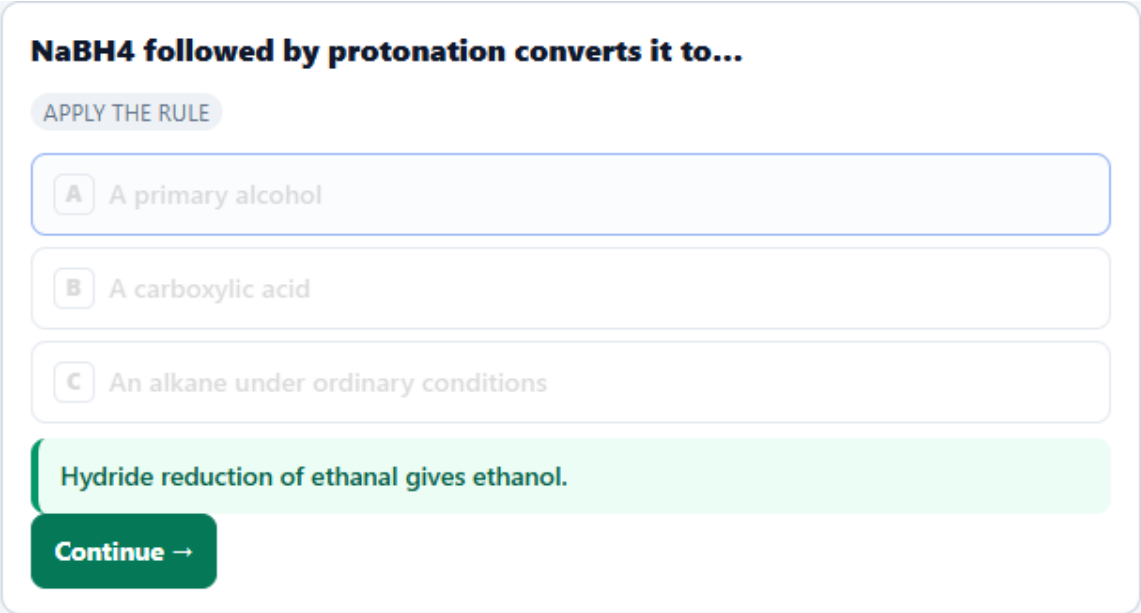
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hydride adds to C=O.

Hint 2: The oxygen remains in the alcohol product.

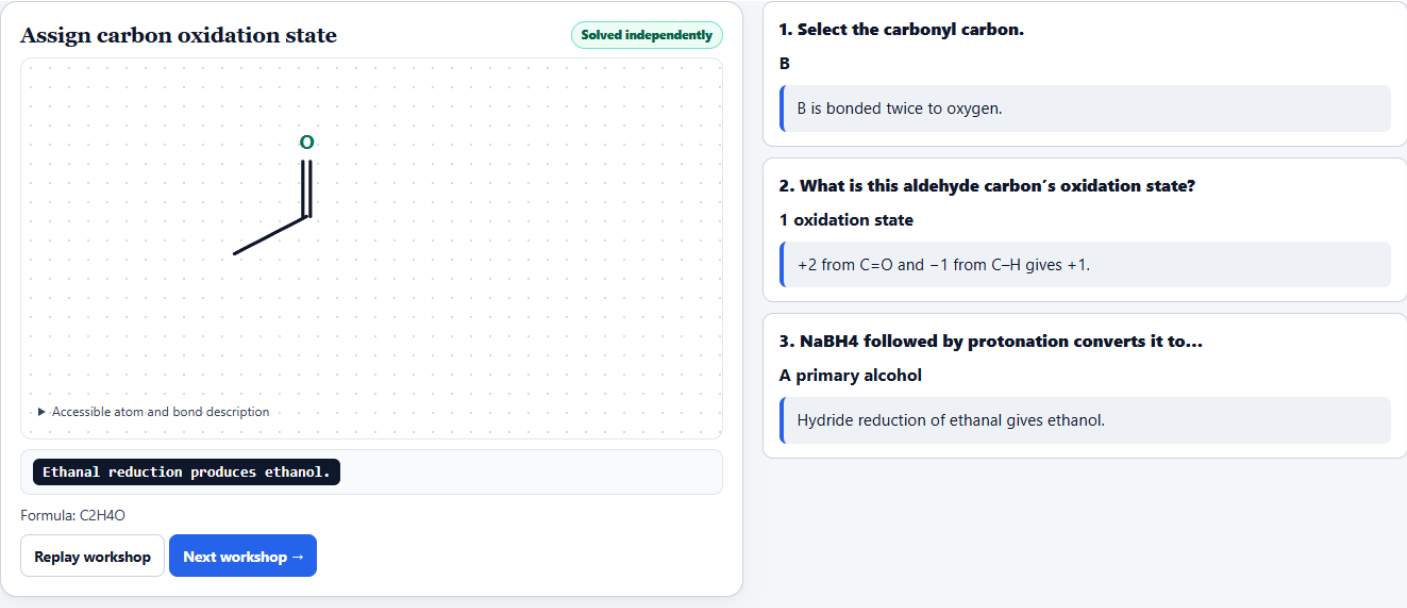
Explanation: Hydride reduction of ethanal gives ethanol.

Screenshot: carbon-redox-stage-3-feedback



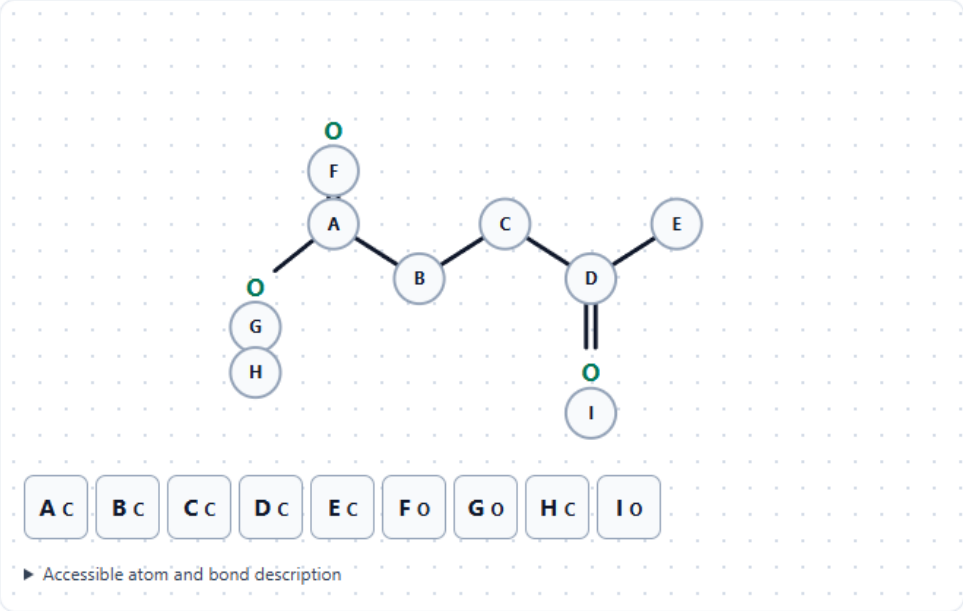
Completion summary: Ethanal reduction produces ethanol.

Screenshot: carbon-redox-complete



selective-reduction | Select a reactive site

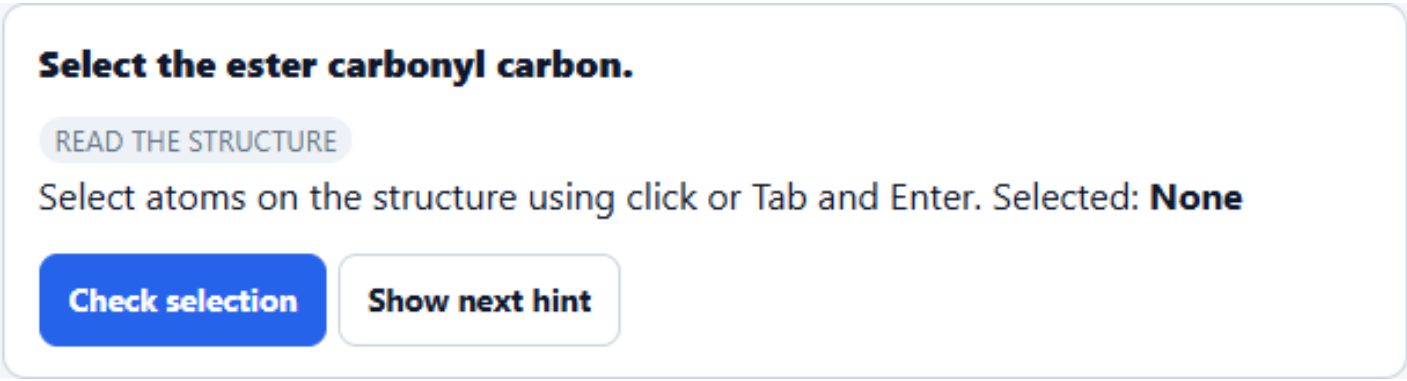
Screenshot: selective-reduction-stage-1-molecule



selective-reduction:stage-1 | Select the ester carbonyl carbon.

Answer type: atoms

Screenshot: selective-reduction-stage-1-question



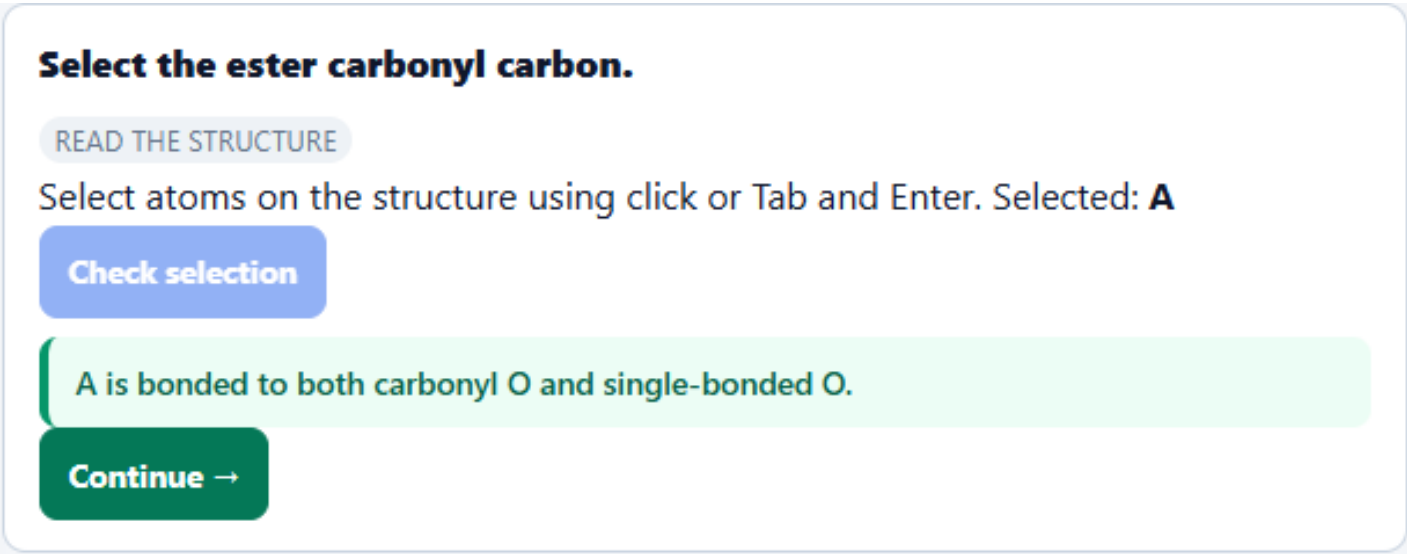
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 5, 6]}}

Hint 1: An ester has C(=O)–O–C.

Hint 2: Find the carbonyl with a second oxygen neighbor.

Explanation: A is bonded to both carbonyl O and single-bonded O.

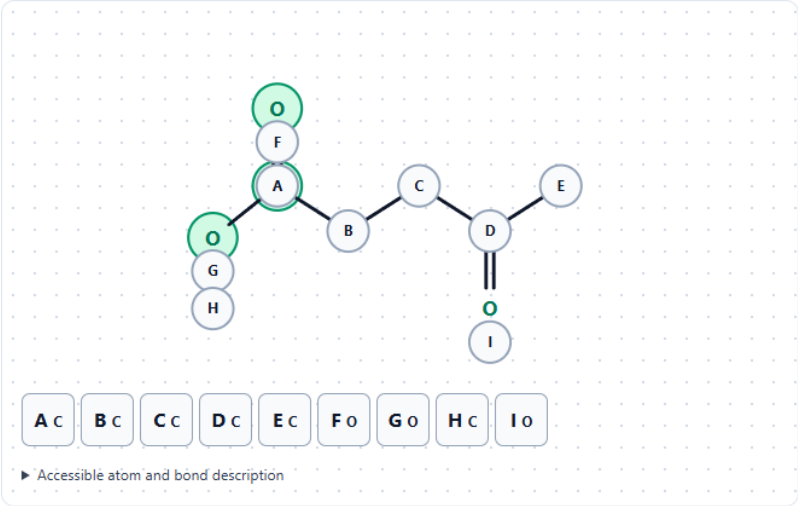
Screenshot: selective-reduction-stage-1-feedback



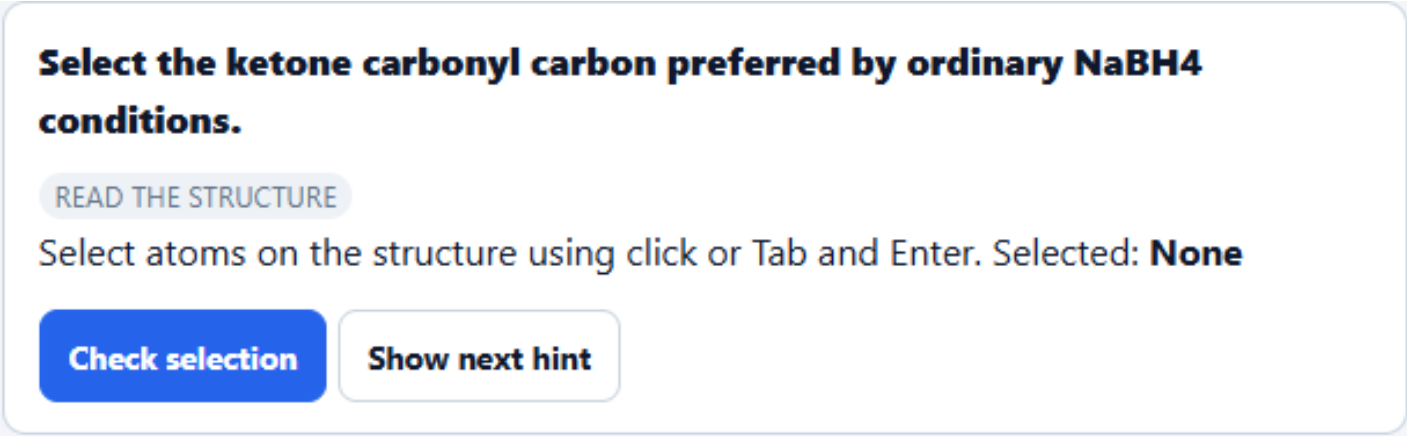
selective-reduction:stage-2 | Select the ketone carbonyl carbon preferred by ordinary NaBH4 conditions.

Answer type: atoms

Screenshot: selective-reduction-stage-2-molecule



Screenshot: selective-reduction-stage-2-question



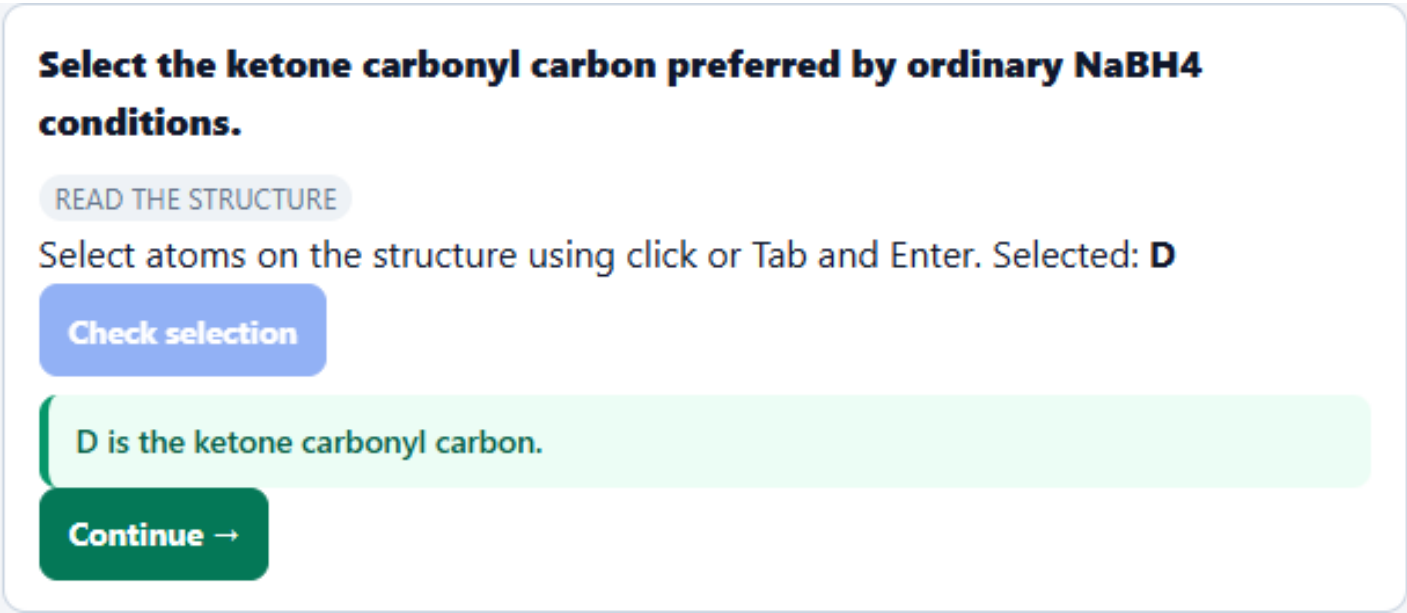
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [3, 8]}}

Hint 1: The ketone has carbon substituents on both sides.

Hint 2: The right-hand carbonyl is more readily reduced here.

Explanation: D is the ketone carbonyl carbon.

Screenshot: selective-reduction-stage-2-feedback



selective-reduction:stage-3 | Why not simply select the highest-priority naming group?

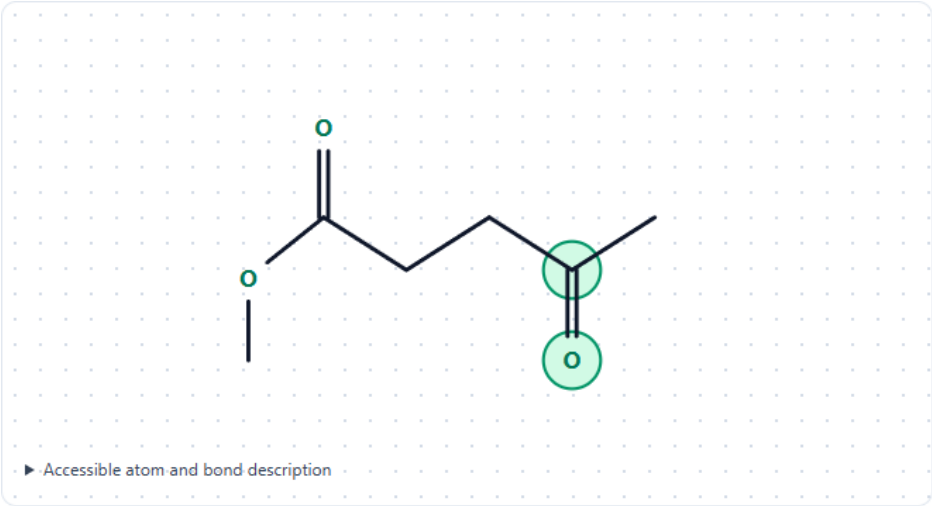
Answer type: choice

Option 0 (A): Naming seniority does not determine kinetic reactivity

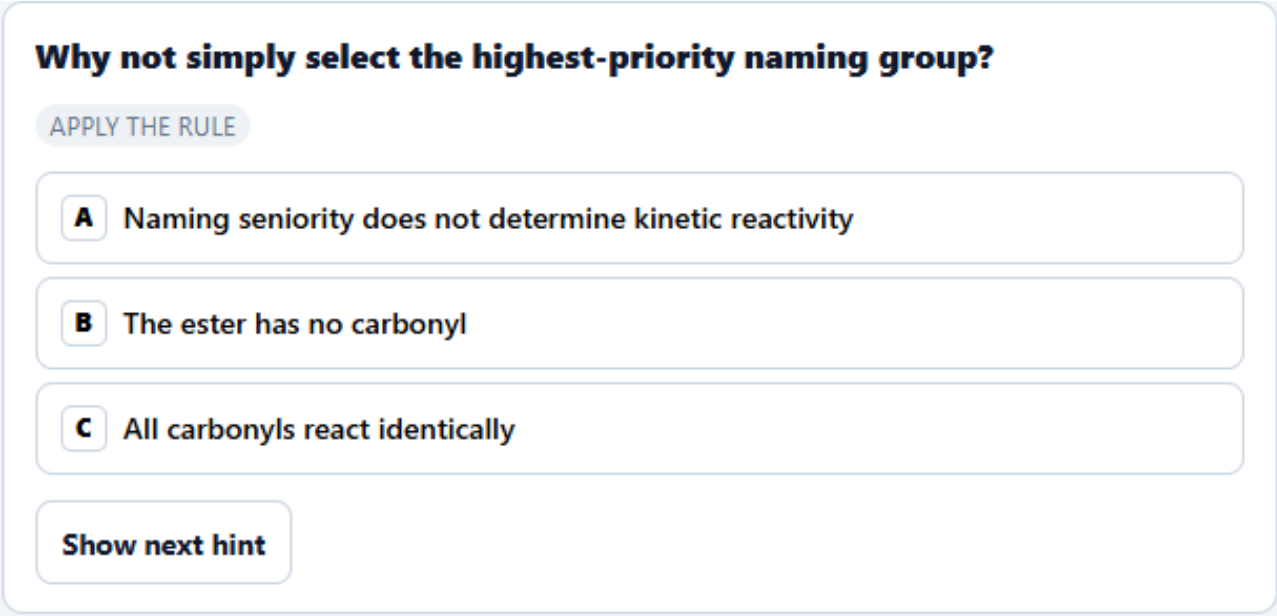
Option 1 (B): The ester has no carbonyl

Option 2 (C): All carbonyls react identically

Screenshot: selective-reduction-stage-3-molecule



Screenshot: selective-reduction-stage-3-question



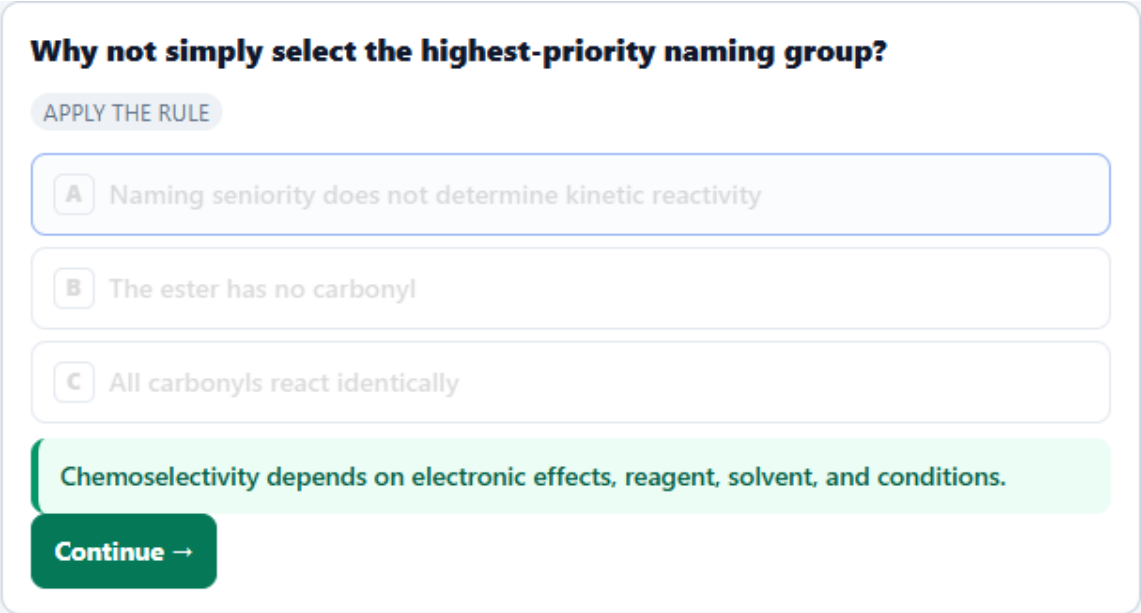
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Naming and reaction mechanisms solve different problems.

Hint 2: Compare resonance donation and electrophilicity.

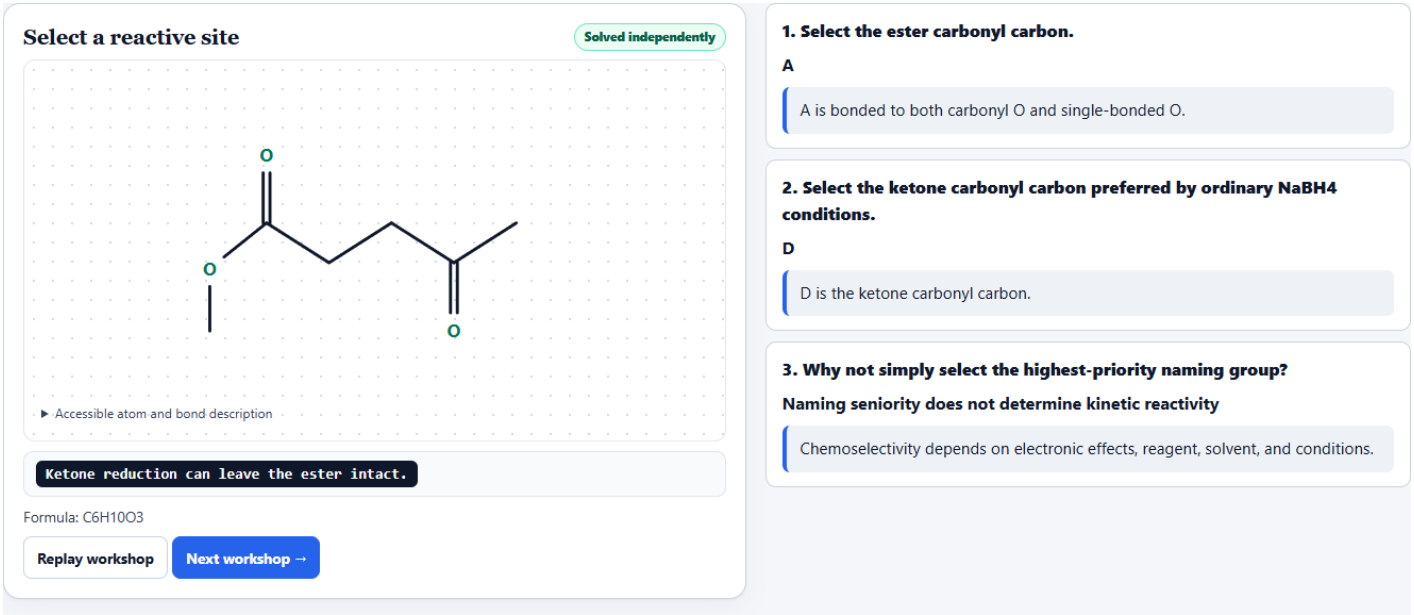
Explanation: Chemoselectivity depends on electronic effects, reagent, solvent, and conditions.

Screenshot: selective-reduction-stage-3-feedback



Completion summary: Ketone reduction can leave the ester intact.

Screenshot: selective-reduction-complete



Molecular repository: author claims to verify

aceticacid

Author claims: ethanoic acid | formula C2H4O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

bromoethane

Author claims: bromoethane | formula C2H5Br

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): Br; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

tbutylbromide

Author claims: 2-bromo-2-methylpropane | formula C4H9Br

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=25.00; charge=0; explicit H override=none

4 (E): Br; x=108.00, y=155.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

ethanol

Author claims: ethanol | formula C2H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

acetaldehyde

Author claims: ethanal | formula C2H4O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

acetone

Author claims: propanone | formula C3H6O

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

ketoester

Author claims: methyl 4-oxopentanoate | formula C6H10O3

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=70.00, y=100.00; charge=0; explicit H override=none

1 (B): C; x=125.00, y=135.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=235.00, y=135.00; charge=0; explicit H override=none

4 (E): C; x=290.00, y=100.00; charge=0; explicit H override=none

5 (F): O; x=70.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=20.00, y=140.00; charge=0; explicit H override=none

7 (H): C; x=20.00, y=195.00; charge=0; explicit H override=none

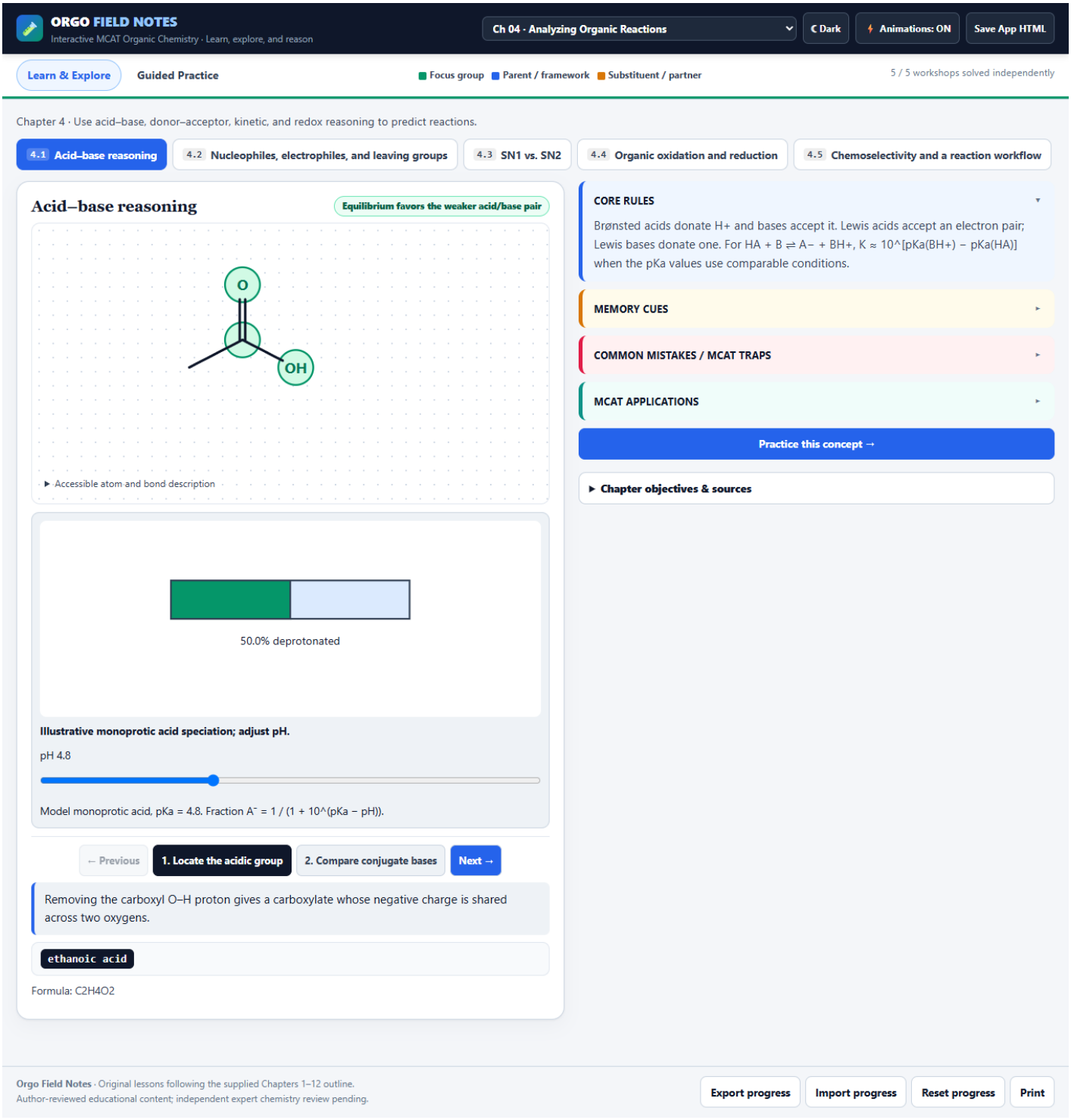
8 (I): O; x=235.00, y=195.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [0, 5, 2], [0, 6, 1], [6, 7, 1], [3, 8, 2]]

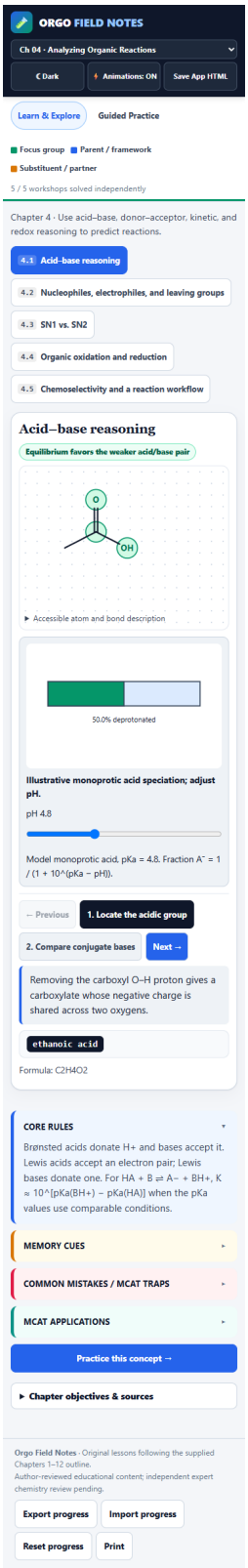
Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

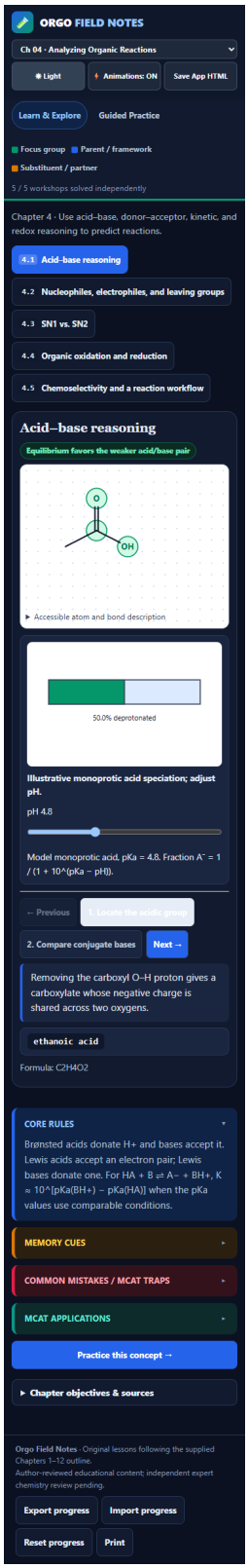
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark



Execution evidence

67 screenshots and 17 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.